

YASKAWA DX100 CONTROLLER INSTRUCTIONS MANUAL



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Other ManualsLib Projects

DX100 OPTIONS INSTRUCTIONS

FOR HIGH-SPEED ETHERNET SERVER FUNCTION

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

MOTOMAN INSTRUCTIONS

MOTOMAN-□□□ INSTRUCTIONS
DX100 INSTRUCTIONS
DX100 OPERATOR'S MANUAL
DX100 MAINTENANCE MANUAL

The DX100 Operator's manual above corresponds to specific usage.
Be sure to use the appropriate manual.

Part Number: 162529-1CD
Revision: 5

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Printed in the United States of America

First Printing, 2012

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MANDATORY

- This manual explains the high-speed Ethernet server function of the DX100 system and general operations. Read this manual carefully and be sure to understand its contents before handling the DX100.
- General items related to safety are listed in Chapter 1: Safety of the DX100 Instructions. To ensure correct and safe operation, carefully read the DX100 Instructions before reading this manual.



CAUTION

- Some drawings in this manual are shown with the protective covers or shields removed for clarity. Be sure all covers and shields are replaced before operating this product.
- The drawings and photos in this manual are representative examples and differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice when necessary due to product improvements, modifications, or changes in specifications.
- If such modification is made, the manual number will also be revised.
- If your copy of the manual is damaged or lost, contact a YASKAWA representative to order a new copy. The representatives are listed on the back cover. Be sure to tell the representative the manual number listed on the front cover.
- YASKAWA is not responsible for incidents arising from unauthorized modification of its products. Unauthorized modification voids your product's warranty.

We suggest that you obtain and review a copy of the ANSI/RIA National Safety Standard for Industrial Robots and Robot Systems (ANSI/RIA R15.06-2012). You can obtain this document from the Robotic Industries Association (RIA) at the following address:

Robotic Industries Association
900 Victors Way
P.O. Box 3724
Ann Arbor, Michigan 48106
TEL: (734) 994-6088
FAX: (734) 994-3338
www.roboticsonline.com

Ultimately, well-trained personnel are the best safeguard against accidents and damage that can result from improper operation of the equipment. The customer is responsible for providing adequately trained personnel to operate, program, and maintain the equipment. **NEVER ALLOW UNTRAINED PERSONNEL TO OPERATE, PROGRAM, OR REPAIR THE EQUIPMENT!**

We recommend approved Yaskawa training courses for all personnel involved with the operation, programming, or repair of the equipment.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

Notes for Safe Operation

Read this manual carefully before installation, operation, maintenance, or inspection of the DX100.

In this manual, the Notes for Safe Operation are classified as “DANGER”, “WARNING”, “CAUTION”, “MANDATORY” or “PROHIBITED”.



DANGER

Indicates an imminent hazardous situation which, if not avoided, could result in death or serious injury to personnel.



WARNING

Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury to personnel.



CAUTION

Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury to personnel and damage to equipment. It may also be used to alert against unsafe practices.



MANDATORY

Always be sure to follow explicitly the items listed under this heading.



PROHIBITED

Must never be performed.

Even items described as “CAUTION” may result in a serious accident in some situations.

At any rate, be sure to follow these important items



To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” and “CAUTION”.



WARNING

- Before operating the manipulator, check that servo power is turned OFF pressing the emergency stop buttons on the programming pendant.

When the servo power is turned OFF, the SERVO ON LED on the programming pendant is turned OFF.

Injury or damage to machinery may result if the emergency stop circuit cannot stop the manipulator during an emergency. The manipulator should not be used if the emergency stop buttons do not function.

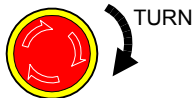
Figure 1: Emergency Stop Button



- Once the emergency stop button is released, clear the cell of all items which could interfere with the operation of the manipulator. Then turn the servo power ON.

Injury may result from unintentional or unexpected manipulator motion.

Figure 2: Release of Emergency Stop



- Observe the following precautions when performing teaching operations within the P-point maximum envelope of the manipulator:
 - Be sure to use a lockout device to the safeguarding when going inside. Also, display the sign that the operation is being performed inside the safeguarding and make sure no one closes the safeguarding. View the manipulator from the front whenever possible.
 - Always follow the predetermined operating procedure.
 - Keep in mind the emergency response measures against the manipulator's unexpected motion toward you.
 - Ensure that you have a safe place to retreat in case of emergency.

Improper or unintended manipulator operation may result in injury.

- Confirm that no person is present in the P-point maximum envelope of the manipulator and that you are in a safe location before:
 - Turning ON the power for the DX100.
 - Moving the manipulator with the programming pendant.
 - Running the system in the check mode.
 - Performing automatic operations.

Injury may result if anyone enters the P-point maximum envelope of the manipulator during operation. Always press an emergency stop button immediately if there is a problem.

The emergency stop buttons is on the programming pendant.



CAUTION

- Perform the following inspection procedures prior to conducting manipulator teaching. If problems are found, repair them immediately, and be sure that all other necessary processing has been performed.
 - Check for problems in manipulator movement.
 - Check for damage to insulation and sheathing of external wires.
- Always return the programming pendant to the hook on the cabinet of the DX100 after use.

The programming pendant can be damaged if it is left in the manipulator's work area, on the floor, or near fixtures.

- Read and understand the Explanation of Warning Labels in the DX100 Instructions before operating the manipulator:

Definition of Terms Used In this Manual

The MOTOMAN is the YASKAWA industrial robot product.

The MOTOMAN usually consists of the manipulator, the controller, the programming pendant, and the manipulator cables.

In this manual, the equipment is designated as follows:

Equipment	Manual Designation
DX100 controller	DX100
DX100 programming pendant	Programming pendant
Cable between the manipulator and the controller	Manipulator cable

Descriptions of the programming pendant, buttons, and displays are shown as follows:

Equipment		Manual Designation
Programming Pendant	Character Keys	The keys which have characters printed on them are denoted with []. ex. [ENTER]
	Symbol Keys	The keys which have a symbol printed on them are not denoted with [] but depicted with a small picture. ex. page key  The cursor key is an exception, and a picture is not shown.
	Axis Keys Number Keys	“Axis Keys” and “Number Keys” are generic names for the keys for axis operation and number input.
	Keys pressed simultaneously	When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, ex. [SHIFT]+[COORD]
	Displays	The menu displayed in the programming pendant is denoted with { }. ex. {JOB}

Description of the Operation Procedure

In the explanation of the operation procedure, the expression [Select •••]” means that the cursor is moved to the object item and the SELECT key is pressed, or that the item is directly selected by touching the screen.

Registered Trademark

In this manual, names of companies, corporations, or products are trademarks, registered trademarks, or brand names for each company or corporation. The indications of (R) and TM are omitted.

Safeguarding Tips

All operators, programmers, maintenance personnel, supervisors, and anyone working near the system must become familiar with the operation of this equipment. All personnel involved with the operation of the equipment must understand potential dangers of operation. General safeguarding tips are as follows:

- Improper operation can result in personal injury and/or damage to the equipment. Only trained personnel familiar with the operation of this equipment, the operator's manuals, the system equipment, and options and accessories should be permitted to operate this equipment.
- Improper connections can damage the equipment. All connections must be made within the standard voltage and current ratings of the equipment.
- The system must be placed in Emergency Stop (E-Stop) mode whenever it is not in use.
- In accordance with ANSI/RIA R15.06-2012, section 4.2.5, Sources of Energy, use lockout/tagout procedures during equipment maintenance. Refer also to Section 1910.147 (29CFR, Part 1910), Occupational Safety and Health Standards for General Industry (OSHA).

Mechanical Safety Devices

The safe operation of this equipment is ultimately the users responsibility. The conditions under which the equipment will be operated safely should be reviewed by the user. The user must be aware of the various national codes, ANSI/RIA R15.06-2012 safety standards, and other local codes that may pertain to the installation and use of this equipment.

Additional safety measures for personnel and equipment may be required depending on system installation, operation, and/or location. The following safety equipment is provided as standard:

- Safety barriers
- Door interlocks
- Emergency stop palm buttons located on operator station

Check all safety equipment frequently for proper operation. Repair or replace any non-functioning safety equipment immediately.

Programming, Operation, and Maintenance Safety

All operators, programmers, maintenance personnel, supervisors, and anyone working near the system must become familiar with the operation of this equipment. Improper operation can result in personal injury and/or damage to the equipment. Only trained personnel familiar with the operation, manuals, electrical design, and equipment interconnections of this equipment should be permitted to program, or maintain the system. All personnel involved with the operation of the equipment must understand potential dangers of operation.

- Inspect the equipment to be sure no potentially hazardous conditions exist. Be sure the area is clean and free of water, oil, debris, etc.
- Be sure that all safeguards are in place. Check all safety equipment for proper operation. Repair or replace any non-functioning safety equipment immediately.
- Check the E-Stop button on the operator station for proper operation before programming. The equipment must be placed in Emergency Stop (E-Stop) mode whenever it is not in use.
- Back up all programs and jobs onto suitable media before program changes are made. To avoid loss of information, programs, or jobs, a backup must always be made before any service procedures are done and before any changes are made to options, accessories, or equipment.
- Any modifications to the controller unit can cause severe personal injury or death, as well as damage to the robot! Do not make any modifications to the controller unit. Making any changes without the written permission from Yaskawa will void the warranty.
- Some operations require standard passwords and some require special passwords.
- The equipment allows modifications of the software for maximum performance. Care must be taken when making these modifications. All modifications made to the software will change the way the equipment operates and can cause severe personal injury or death, as well as damage parts of the system. Double check all modifications under every mode of operation to ensure that the changes have not created hazards or dangerous situations.
- This equipment has multiple sources of electrical supply. Electrical interconnections are made between the controller and other equipment. Disconnect and lockout/tagout all electrical circuits before making any modifications or connections.
- Do not perform any maintenance procedures before reading and understanding the proper procedures in the appropriate manual.
- Use proper replacement parts.
- Improper connections can damage the equipment. All connections must be made within the standard voltage and current ratings of the equipment.

Maintenance Safety

Turn the power OFF and disconnect and lockout/tagout all electrical circuits before making any modifications or connections.

Perform only the maintenance described in this manual. Maintenance other than specified in this manual should be performed only by Yaskawa-trained, qualified personnel.

Summary of Warning Information

This manual is provided to help users establish safe conditions for operating the equipment. Specific considerations and precautions are also described in the manual, but appear in the form of Dangers, Warnings, Cautions, and Notes.

It is important that users operate the equipment in accordance with this instruction manual and any additional information which may be provided by Yaskawa. Address any questions regarding the safe and proper operation of the equipment to Yaskawa Customer Support.

Customer Support Information

If you need assistance with any aspect of your DX100 system, please contact YASKAWA Customer Support at the following 24-hour telephone number:

(937) 847-3200

For **routine** technical inquiries, you can also contact YASKAWA Customer Support at the following e-mail address:

techsupport@motoman.com

When using e-mail to contact YASKAWA Customer Support, please provide a detailed description of your issue, along with complete contact information. Please allow approximately 24 to 36 hours for a response to your inquiry.



Please use e-mail for **routine** inquiries only. If you have an urgent or emergency need for service, replacement parts, or information, you must contact YASKAWA Customer Support at the telephone number shown above.

Please have the following information ready before you call Customer Support:

- | | |
|----------------------------|--|
| • System | DX100 |
| • Robots | _____ |
| • Primary Application | _____ |
| • Controller | DX100 |
| • Software Version | Access this information on the Programming Pendant's LCD display screen by selecting {MAIN MENU} - {SYSTEM INFO} - {VERSION} |
| • Robot Serial Number | Located on the robot data plate |
| • Robot Sales Order Number | Located on the DX100 controller data plate |

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1 Introductions

The high-speed Ethernet server function is a new communication protocol to enable high-speed Ethernet communication between the DX100 and external devices such as PC, etc.

Followings are the characteristics of this function.

- (1) It becomes possible to communicate in more than two times higher speed than the present Ethernet server function and more than 5 times higher speed than the present Ethernet data transmission function.
- (2) It combines the present Ethernet data transmission function (host control) and the present Ethernet server function. (except for some functions)
- (3) It corresponds to the file receiving/transmission function to which the present Ethernet server function does not correspond.
- (4) It is incompatible to the present data transmission function (host control) and the present Ethernet server function. Therefore, MotoCom communication library (Ver3.6), which corresponds to the high-speed Ethernet server function, will be released at the same time.
- (5) It is also possible to create a communication program without using MotoCom since this function is publishing its communication protocol.
- (6) To maintain the compatibility with existing communication software, the present data transmission function and the present Ethernet server function are still available.

1.1 Preparation

This high-speed Ethernet server function is an expansion option to the DX100 Ethernet function. In this reason, when using this function, the PC should be ready to use the DX100 Ethernet function.

1.2 Restriction

- This function cannot use concurrently with MotoPlus function, PP customizing function, other Ethernet functions and the data transmission function (serial).
- To increase the speed, the protocol of this function was modified. Therefore, it has no compatibility with the data transmission function and the Ethernet server function.
To retain the compatibility, MotoCom communication library (Ver3.6) will be released at the same time with this function. Please use MotoCom communication library of later version than Ver3.6.

2 System Setting

To use the high-speed Ethernet server function, configuration of the following settings are required.

2.1 Before using the System

The high-speed Ethernet server function is designed as an expansion option to the DX100 Ethernet function. Before using this function, it is required to make the DX100 Ethernet host control function available.

For more details, see chapter 3 “Ethernet Function Settings” in the “DX100 OPTIONS INSTRUCTIONS FOR ETHERNET FUNCTION”.

2.2 Parameter Setting

Set the following parameters before using this function.

Parameter	Details	Setting value
RS022	Instance 0 permitted (Instance 0 is used as the ordinal data)	1
RS029	A job during the playback operation, Loading of a variable	1
RS034	Timer to wait for a replay	200
RS035	Timer for monitoring end of text	200

2.3 Setting of Relevant Parameter

Parameter	Details	When shipping
S2C541	Specify the permission of variable and I/O input during the play mode (0: writing is allowed / 1: writing is not allowed)	1
S2C542	Specify the permission of variable and I/O input during the edit-lock status (0: writing is allowed / 1: writing is not allowed)	1
S2C680	Specify the permission of the batch data backup function (0: INVALID 1: Create RAMDISK at the STARTUP)	1



When setting 0 to S2C541 (writing is allowed), writing is possible even during the playback operation. However, please be noted that this setting may affect the manipulator's cycle time due to some writing timings or their frequencies.

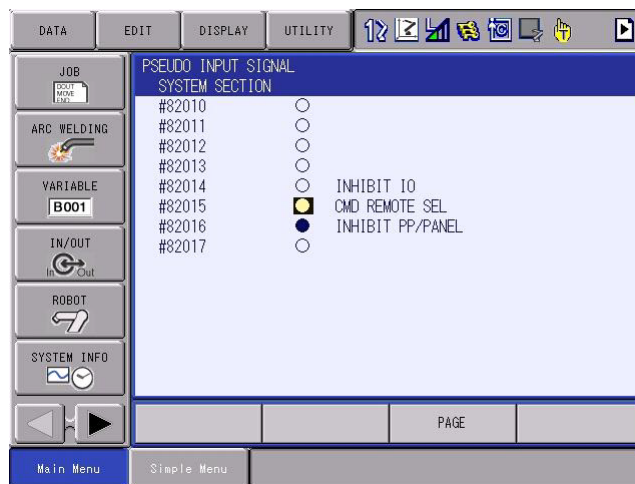


Following are the status to which specifying of the “edit-lock status” is permitted by S2C542 parameter.

- During an alarm
- When an external memory device is operated
- When the data transmission function is used
- Specific input EDIT_LOCK (#40064) is turned ON

2.4 Setting of Command Remote

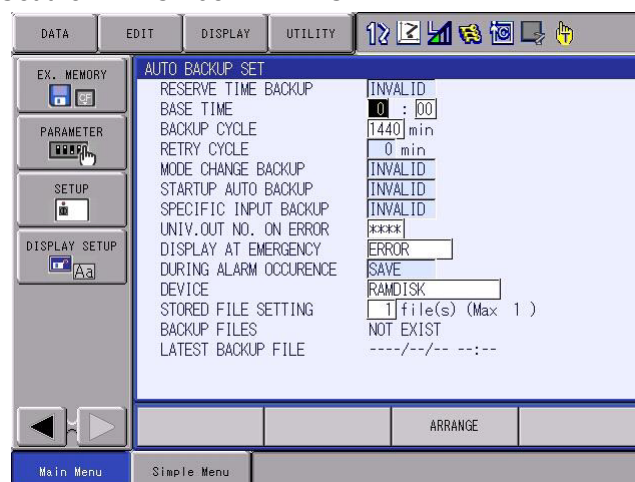
Set Management mode as Security mode, and select {IN/OUT} – {PSEUDO INPUT SIGNAL} to appear the following display. Move the cursor to the #82015 CMD REMOTE SEL, and press [INTER LOCK] + [SELECT] to select [ON].



2.5 Setting of a Batch Data Backup Function

With the batch data backup function, the data saved in the DX100 such as system setting or operational condition are collectively backed up by using the command from High Speed Ethernet Server Function. Set the following procedures in advance to use this function.

Set Management mode as Security mode. Select {CONTROLLER SET} + {AUTO BACKUP SET} in the Main Menu, and following display will appear. Set the DEVICE as "RAMDISK".





- A batch data backup function is available for software version DS3.00 or higher.
- Refer to DX100 Instruction section 9.3 “Auto Backup Function” for more details.
- Refer to DX100 Instruction section 9.4 “Loading the Backup Data from the CompactFlash” for using files from the restore system which is backed up by command from High Speed Ethernet Server Function.
- During an alarm is occurring, it would not be able to change the device. Thus, operate after resetting the alarm.
- When the parameter is S2C680=0, “RAMDISK” will not appear in the “AUTO BACKUP SET” display. Make sure to set the parameter S2C680.

3 Transmission Procedure

3.1 Packet Format

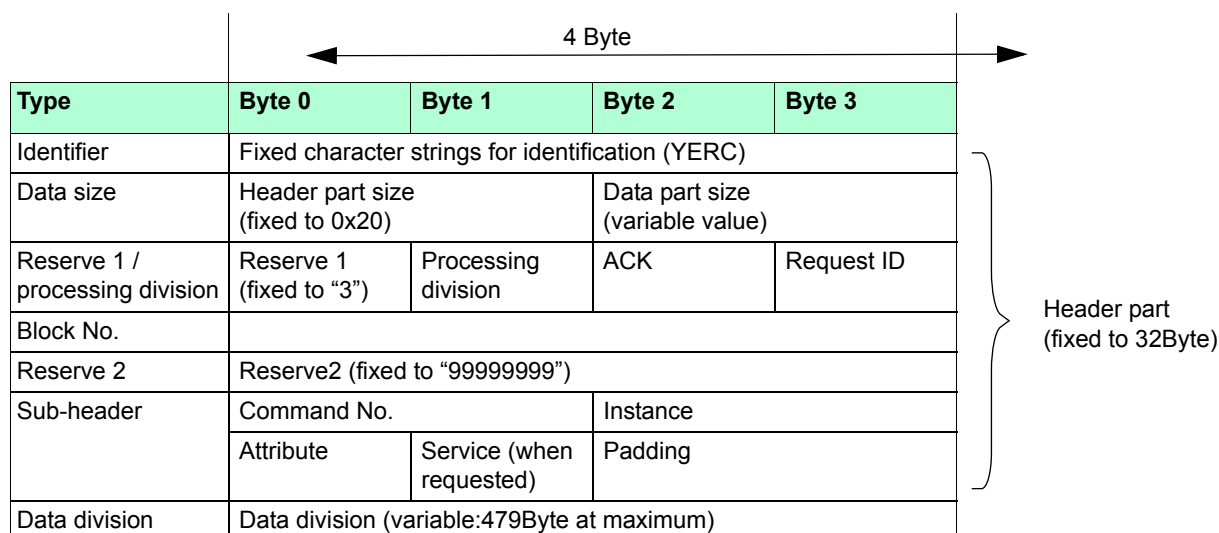
Transmission packet of the high-speed Ethernet server function is composed of header part (32 byte) + data part (changeable: 479 byte at max.)

The transmission packet consists of “request”, which transmits the data from the PC to the DX100, and “answer”, which transmits the data from the DX100 to the PC.

The sub-header setting composition of “request” and “answer” are different. And the setting value of the “answer” varies in accordance with the replying contents.

Followings are the format of each packet.

Request (the PC to the DX100)



Answer (the DX100 to the PC)

	4 Byte			
Type	Byte 0	Byte 1	Byte 2	Byte 3
Identifier	Fixed character strings for identification (YERC)			
Data size	Header part size (fixed to 0x20)		Data part size (variable value)	
Reserve 1 / processing division	Reserve 1 (fixed to “3”)	Processing division	ACK	Request ID
Block No.	Allocate the block number from 0 to 0x7fff_ffff Add 0x8000_0000 to the last block			
Reserve 2	Reserve 2 (fixed to “99999999”)			
Sub-header	Service (when replying)	Status: When normal operation: 0x00 When abnormal operation: other than 0x00	Added status size	Padding
			Added status size ¹⁾ Padding	
Data division	Data division (variable: 479Byte at maximum)			

Header part
(fixed to 32Byte)

1 For the details of added status, please refer to *chapter 5 "Added Status Code"*.

Item		Data size	Settings
Identifier		4byte	Fixed to "YERC"
Header part size		2byte	Size of header part (fixed to 0x20)
Data part size		2byte	Size of data part (variable)
Reserve 1		1byte	Fixed to "3"
Processing division		1byte	1: robot control 2: file control
ACK		1byte	Request: 0 Other than request: 1
Request ID		1byte	Identifying ID for command session (increment this ID every time the client side outputs a new command. In reply to this, server side answers the received value.)
Block No.		4byte	Request: 0 Answer: add 0x8000_0000 to the last packet. Data transmission other than above: add 1 (max: 0x7FFF_FFFF)
Reserve 2		8byte	Fixed to "99999999"
Sub-header (request)	Command No.	2byte	Execute processing by this command. (conforms to "Class" of CIP communication protocol)
	Instance	2byte	Define SECTION to execute a command. (conforms to "Padding" of CIP communication protocol)
	Attribute	1byte	Define SUB SECTION for executing a command. Attribute: (conforms to "Attribute" of CIP communication protocol)
	Service (request)	1byte	Define data accessing method.
Sub-header (answer)	Service (answer)	1byte	Add 0s80 to service (request).
	Status	1byte	0x00: normal reply Other than 0x00: abnormal reply
	Added status size	1byte	Size of added status (0: not specified / 1: 1 WORD data / 2: 2 WORD data)
	Added status	2byte	Error code specified by added status size For details, refer to <i>chapter 5 "Added Status Code"</i> .
Padding		Variable	Reserve area

Details of sub-header

- Sub header (request)

Sub header (request)	Command No.		Instance
	Attribute	Service (request)	Padding

- Sub header (answer/ no added status)

Sub header (request)	Service (answer)	Status: normal: 0x00	Added status: size: 0x00	Padding
	Added status:0x14809580		Padding	

- Sub header (answer/ with added status)

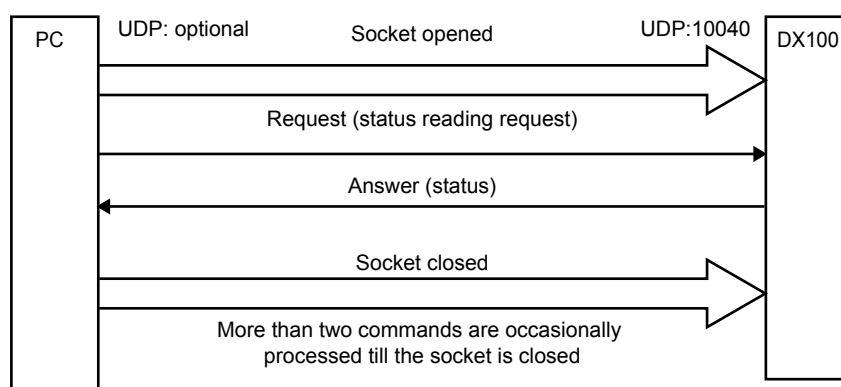
Sub header (request)	Service (answer)	Status: abnormal: other than 0x00	Added status: size:0x01	Padding
	Added status:0x00001010		Padding	

3.2 Outline

The transmission/receiving flow of the transmission packet is divided into robot control and file control. Please refer to *section 3.3 “Respective Commands for Robot Control”* for the details of respective robot control commands (request/answer) and *section 3.4 “File Control Command”* for the details of respective file control commands.

[Ex. When Reading]

3.2.1 Robot Control/Status Reading



Request

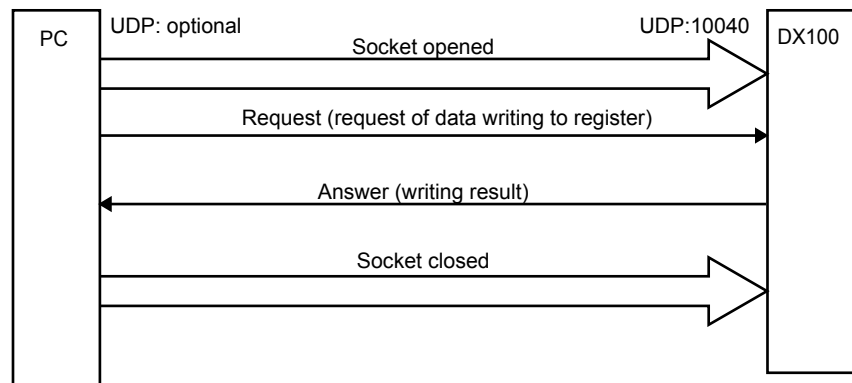
<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x01	0x00	0x00	Reserve 1	Processing division	ACK	Request ID
0x0000_0000				Block No.			
“99999999”				Reserve 2			
0x0072		0x0001		Command No.		Instance	
0x00	0x01	0x0000		Attribute	Service	Padding	

Answer

<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x01	0x01	0x00	Reserve 1	Processing division	ACK	Request ID
0x8000_0000				Block No.			
“99999999”				Reserve 2			
0x81	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	
Status data 1				Reading value 1			
Status data 2				Reading value 2			

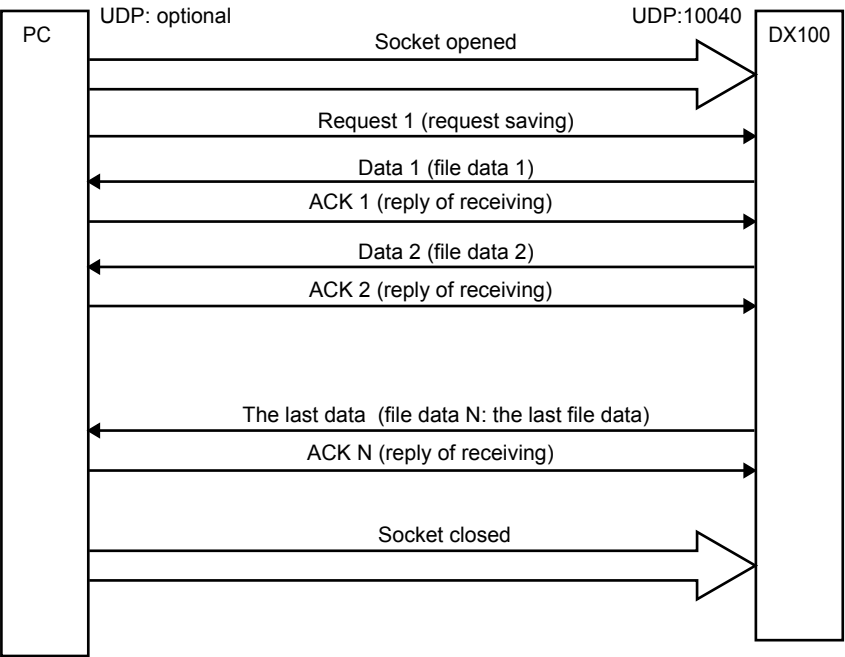
[Ex. When Writing]**3.2.2 Robot Control/Data Writing to Register****Request****<Format>**

“YERC”				Identifier			
0x0020		0x0002		Header part size		Data part size	
0x03	0x01	0x00	0x01	Reserve 1	Processing division	ACK	Request ID
0x0000_0000				Block No.			
‘99999999’				Reserve 2			
0x0079		Register No.		Command No.		Instance	
0x00	0x02	0x0000		Attribute	Service	Padding	
Register data				Writing value			

Answer**<Format>**

'YERC'				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x01	0x01	0x01	Reserve 1	Processing division	ACK	Request ID
0x8000_0000				Block No.			
'99999999'				Reserve 2			
0x82	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	

3.2.3 File Control



Request 1

<Format>

“YERC”				Identifier			
0x0020		0x000B		Header part size		Data part size	
0x03	0x02	0x00	0x02	Reserve 1	Processing division	ACK	Request ID
0x0000_0000				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x16	0x00		Attribute	Service	Padding	
T	E	S	T	File name			
J	O	B	.				
J	B	I					

Data 1

<Format>

“YERC”				Identifier			
0x0020		0x01d f		Header part size		Data part size	
0x03	0x02	0x01	0x02	Reserve 1	Processing division	ACK	Request ID
0x0000_0001				Block No.			
“99999999”				Reserve 2			
0x96	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	
File data 1				File data 1			

ACK1

<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x02	Reserve 1	Processing division	ACK	Request ID
0x0000_0001				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x16	0x00		Attribute	Service	Padding	

Data 2

<Format>

“YERC”				Identifier			
0x0020		0x01d?		Header part size		Data part size	
0x03	0x02	0x01	0x02	Reserve 1	Processing division	ACK	Request ID
0x0000_0002				Block No.			
“99999999”				Reserve 2			
0x96	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	
File data 2				File data 2			

DX100

3 Transmission Procedure

3.2 Outline

ACK2

<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x02	Reserve 1	Processing division	ACK	Request ID
0x0000_0002				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x16	0x00		Attribute	Service	Padding	

The last data (N)

<Format>

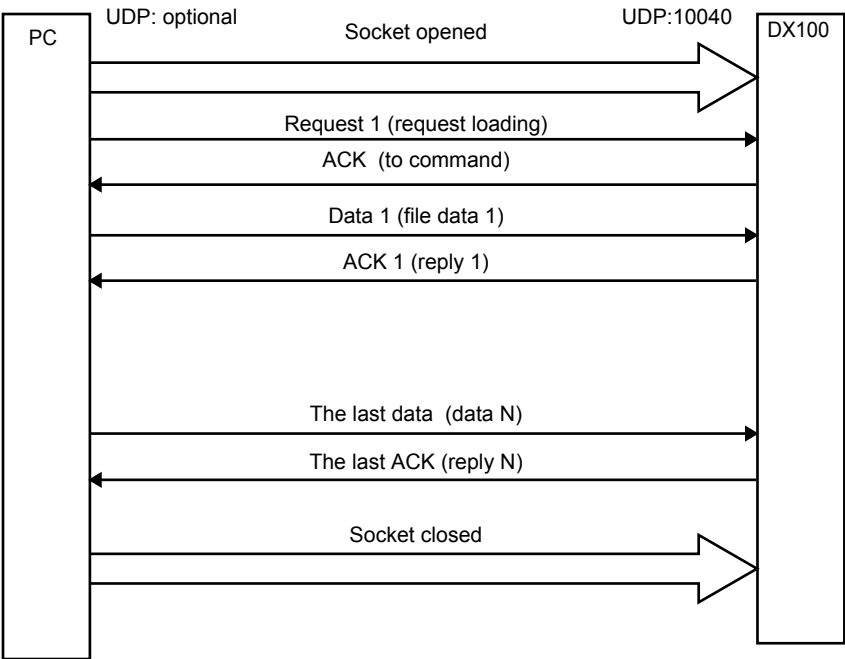
“YERC”				Identifier			
0x0020		0x0008		Header part size		Data part size	
0x03	0x02	0x01	0x02	Reserve 1	Processing division	ACK	Request ID
0x8000_000N				Block No.			
“99999999”				Reserve 2			
0x96	0x00	0x00	0x00	Service	status	Added status size	Padding
0x0000		0x0000		Added status		Padding	
File data N				File data N			

The last ACK (N)

<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x02	Reserve 1	Processing division	ACK	Request ID
0x8000_000N				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x16	0x00		Attribute	Service	Padding	

3.2.4 File Control (File Loading)



Request 1

<Format>

“YERC”				Identifier			
0x0020		0x000B		Header part size		Data part size	
0x03	0x02	0x00	0x03	Reserve 1	Processing division	ACK	Request ID
0x0000_0000				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x0000	0x15	0x00		Attribute	Service	Padding	
T	E	S	T	File name			
J	O	B	.				
J	B	I					

ACK (to request)

<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x03	Reserve 1	Processing division	ACK	Request ID
0x0000_0000				Block No.			
“99999999”				Reserve 2			
0x95	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	

DX100

3 Transmission Procedure

3.2 Outline

Data 1

<Format>

“YERC”				Identifier			
0x0020		0x01d?		Header part size		Data part size	
0x03	0x02	0x01	0x03	Reserve 1	Processing division	ACK	Request ID
0x0000_0001				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x0000	0x15	0x00		Attribute	Service	Padding	
File data 1				File data 1			

ACK1

<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x03	Reserve 1	Processing division	ACK	Request ID
0x0000_0001				Block No.			
“99999999”				Reserve 2			
0x95	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	

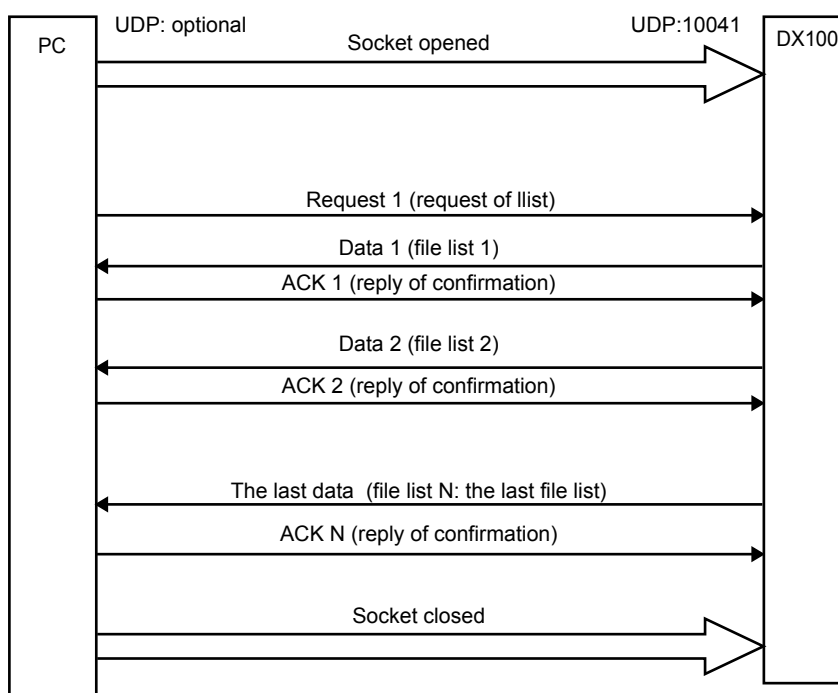
The last data (N)

<Format>

“YERC”				Identifier			
0x0020		0x0008		Header part size		Data part size	
0x03	0x02	0x01	0x03	Reserve 1	Processing division	ACK	Request ID
0x8000_000N				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x0000	0x15	0x00		Attribute	Service	Padding	
File data N				File data N			

The last ACK (N)				<Format>			
“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x03	Reserve 1	Processing division	ACK	Request ID
0x8000_000N				Block No.			
“99999999”				Reserve 2			
0x95	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	

3.2.5 File Control (File list)



Request 1

<Format>

“YERC”				Identifier			
0x0020		0x0005		Header part size		Data part size	
0x03	0x02	0x00	0x04	Reserve 1	Processing division	ACK	Request ID
0x0000_0000				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x32	0x0000		Attribute	Service	Padding	
*	.	J	B	File identification (refer to data details)			
I							

Data 1

<Format>

“YERC”				Identifier			
0x0020		0x01d?		Header part size		Data part size	
0x03	0x02	0x01	0x04	Reserve 1	Processing division	ACK	Request ID
0x0000_0001				Block No.			
“99999999”				Reserve 2			
0xB2	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	
File list 1				File list 1 (refer to “Details of data”)			

ACK1

<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x04	Reserve 1	Processing division	ACK	Request ID
0x0000_0001				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x32	0x0000		Attribute	Service	Padding	

Data 2

<Format>

“YERC”				Identifier			
0x0020		0x01d?		Header part size		Data part size	
0x03	0x02	0x01	0x04	Reserve 1	Processing division	ACK	Request ID
0x0000_0002				Block No.			
“99999999”				Reserve 2			
0xB2	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	
File list 2				File list 2			

ACK2

<Format>

“YERC”				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x04	Reserve 1	Processing division	ACK	Request ID
0x0000_0002				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x32	0x0000		Attribute	Service	Padding	

DX100

3 Transmission Procedure

3.2 Outline

The last data (N)

<Format>

“YERC”“YERC”				Identifier			
0x0020		0x0008		Header part size		Data part size	
0x03	0x02	0x01	0x04	Reserve 1	Processing division	ACK	Request ID
0x8000_000N				Block No.			
“99999999”				Reserve 2			
0xB2	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	
File list N				File list N			

The last ACK (N)

<Format>

"YERC"				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x04	Reserve 1	Processing division	ACK	Request ID
0x8000_000N				Block No.			
"99999999"				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x32	0x0000		Attribute	Service	Padding	

Detail of data

Not specified	JB1 list
.	JB1 list
*.JB1	JB1 list
*.DAT	DAT file list
*.CND	CND file list
*.PRM	PRM file list
*.SYS	SYS file list
*.LST	LST file list

Output form of the list

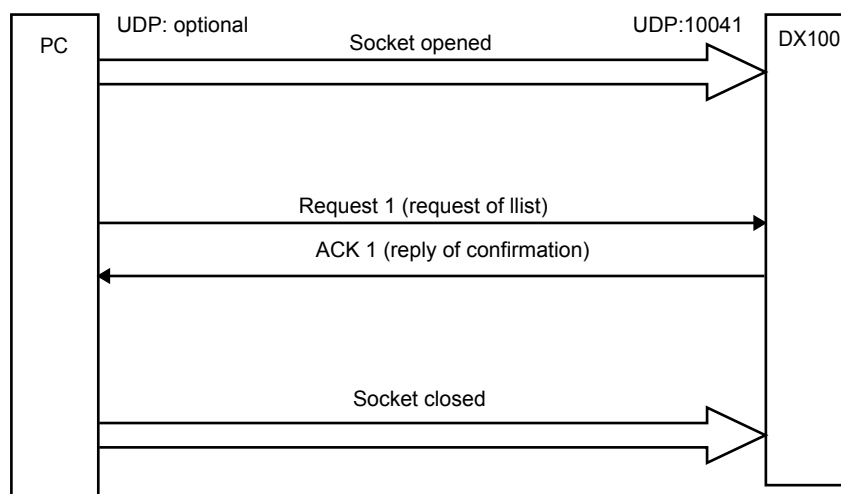
The list is described in the form of “file name” + <CR> + <LF> consecutively

<Ex.>

'1'	'.'	'J'	'B'
'I'	<CR>	<LF>	'2'
'2'	'.'	'J'	'B'
'I'	<CR>	<LF>	'3'
'3'	'3'	'.'	'J'
'B'	'I'	<CR>	<LF>
'4'	'4'	'4'	'4'
'.'	'J'	'B'	'I'
<CR>	<LF>		

<CR><LF> means end-of -line
<CR> : Carriage Return
<LF> : Line Feed

3.2.6 File Control (Deleting of file)



Request 1

<Format>

“YERC”				Identifier			
0x0020		0x000B		Header part size		Data part size	
0x03	0x02	0x01	0x05	Reserve 1	Processing division	ACK	Request ID
0x0000_0000				Block No.			
“99999999”				Reserve 2			
0x00		0x0000		Command No.		Instance	
0x00	0x09	0x00		Attribute	Service	Padding	
T	E	S	T	File name			
J	O	B	.				
J	B	I					

ACK 1

<Format>

'YERC'				Identifier			
0x0020		0x0000		Header part size		Data part size	
0x03	0x02	0x01	0x05	Reserve 1	Processing division	ACK	Request ID
0x8000_0000				Block No.			
"99999999"				Reserve 2			
0x89	0x00	0x00	0x00	Service	Status	Added status size	Padding
0x0000		0x0000		Added status		Padding	

3.3 Respective Commands for Robot Control

Follows are robot controlling commands which can use in the high-speed Ethernet communication.

Table 3-1: List of Robot Control Command

No.	Command No.	Name	Reference chapter
1	0x70	Alarm data reading command	Refer to <i>section 3.3.1.</i>
2	0x71	Alarm history reading command	Refer to <i>section 3.3.2.</i>
3	0x72	Status information reading command	Refer to <i>section 3.3.3.</i>
4	0x73	Executing job information reading command	Refer to <i>section 3.3.4.</i>
5	0x74	Axis configuration information reading command	Refer to <i>section 3.3.5.</i>
6	0x75	Robot position data reading command	Refer to <i>section 3.3.6.</i>
7	0x76	Position error reading command	Refer to <i>section 3.3.7.</i>
8	0x77	Torque data reading command	Refer to <i>section 3.3.8.</i>
9	0x78	I/O data reading / writing command	Refer to <i>section 3.3.9.</i>
10	0x79	Register data reading / writing command	Refer to <i>section 3.3.10.</i>
11	0x7A	Byte variable (B) reading / writing command	Refer to <i>section 3.3.11.</i>
12	0x7B	Integer type variable (I) reading / writing command	Refer to <i>section 3.3.12.</i>
13	0x7C	Double precision integer type variable (B) reading / writing command	Refer to <i>section 3.3.13.</i>
14	0x7D	Real type variable (R) reading / writing command	Refer to <i>section 3.3.14.</i>
15	0x7E	Character type variable (S) reading / writing command	Refer to <i>section 3.3.15.</i>
16	0x7F	Robot position type variable (P) reading / writing command	Refer to <i>section 3.3.16.</i>
17	0x80	Base position type variable (BP) reading / writing command	Refer to <i>section 3.3.17.</i>
18	0x81	External axis type variable (EX) reading / writing command	Refer to <i>section 3.3.18.</i>
19	0x82	Alarm reset / error cancel command	Refer to <i>section 3.3.19.</i>
20	0x83	HOLD / servo ON/OFF command	Refer to <i>section 3.3.20.</i>
21	0x84	Step / cycle / continuous switching command	Refer to <i>section 3.3.21.</i>
22	0x85	Character string display command to the programming pendant	Refer to <i>section 3.3.22.</i>
23	0x86	Start-up (job START) command	Refer to <i>section 3.3.23.</i>
24	0x87	Job select command	Refer to <i>section 3.3.24.</i>
25	0x88	Management time acquiring command	Refer to <i>section 3.3.25.</i>
26	0x89	System information acquiring command	Refer to <i>section 3.3.26.</i>
27	0x300	Plural I/O data reading / writing command	Refer to <i>section 3.3.27.</i>
28	0x301	Plural register data reading / writing command	Refer to <i>section 3.3.28.</i>
29	0x302	Plural byte type variable (B) reading / writing command	Refer to <i>section 3.3.29.</i>
30	0x303	Plural integer type variable (I) reading / writing command	Refer to <i>section 3.3.30.</i>
31	0x304	Plural double precision integer type variable (B) reading / writing command	Refer to <i>section 3.3.31.</i>
32	0x305	Plural real type variable (R) reading / writing command	Refer to <i>section 3.3.32.</i>
33	0x306	Plural character type variable (S) reading / writing command	Refer to <i>section 3.3.33.</i>
34	0x307	Plural robot position type variable (P) reading / writing command	Refer to <i>section 3.3.34.</i>

Table 3-1: List of Robot Control Command

No.	Command No.	Name	Reference chapter
35	0x308	Plural base position type variable (BP) reading / writing command	Refer to <i>section 3.3.35</i> .
36	0x309	Plural external axis type variable (EX) reading / writing command	Refer to <i>section 3.3.36</i> .
37	0x30A	Alarm data reading command (for applying the sub code character strings) ¹⁾	Refer to <i>section 3.3.37</i> .
38	0x30B	Alarm history reading command (for applying the sub character strings) ¹⁾	Refer to <i>section 3.3.38</i> .
39	0x8A	Move instruction command (Type Cartesian coordinates) ¹⁾	Refer to <i>section 3.3.39</i> .
40	0x8B	Move instruction command (Type Pulse) ¹⁾	Refer to <i>section 3.3.40</i> .

¹ This command is available for system software version DS3.00 or higher.

3.3.1 Alarm Data Reading Command

Request

Sub header part

<Details>

Command No.	0x70
Instance	Specify one out of followings 1: The latest alarm 2: The second alarm from the latest 3: The third alarm from the latest 4: The fourth alarm from the latest
Attribute	Specify one out of followings 1: Alarm code 2: Alarm data 3: By alarm type 4: Alarm occurring time 5: Alarm character string name
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01

Up to four alarms are displayed on the P.P display at the same time. Specify one out of them.

Alarm code means the alarm No.
 Alarm data means the sub code which supports the alarm contents. Some alarms may not appear as the sub code.

Specify the accessing method to the data.
 0x0E: Read out data of the specified element number
 0x01: Read out data of all the element number
 (In this case, specify 0 to the element number.)

Data part

No data part

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD
Added status	Error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Alarm code				Range is from 0x0001 to 0x270F(decimal value: 9999)
2	Alarm data				Setting values vary in accordance with the contents of the alarm type. Also, some alarms are not displayed with the sub code. In this case, the value is zero (0x0).
3	Alarm type				0 : No alarm 1 : Decimal UNSIGNED SHORT type (display example: [1]) 2 : UNSIGNED CHAR bit pattern (display example: [0000_0001]) 3 : User axis type (display example: [SLURBT]) 4 : Spacial coordinate type (display example: [XYZ]) 5 : Robot coordinate type (display example: [XYZRxRyRz]) 6 : Conveyor characteristic file (display example: [123]) 8 : Control group type (display example: [R1R2S1S2]) robot & station 9 : Decimal SHORT type (display example: [-1]) 10 : UNSIGNED SHORT bit pattern (display example: [0000_0000_0000_0001]) 11 : Control group type (display example: [R1]) for robot only 12 : Control group type (display example:[R1S1B1]) for robot, station and base 20 : Control group LOW/HIGH logical axis (display example: [R1:LOW SLURBT, HIGH SLURBT]) 21 : Control group MIN/MAX logical axis (display example: [R1: MIN SLURBT, MAX SLURBT]) 22 : Control group MIN/MAX spacial coordinate (display example: [R1: MIN XYZ, MAX XYZ]) 23 : Logical axis of both control group 1 and control group 2 (display example: [R1: SLURBT, R2: SLURBT]) 24 : Logical axis 1 and 2 of the control group (display example: [R1: SLURBT, SLURBT]) 25 : Logical axis of the control group and UNSIGNED CHAR type (display example: [R1: SLURBT, 1]) 27 : Control group and UNSIGNED CHAR type (display example: [R1: 1])
4	Alarm occurring time (Character strings of 16 letters) Ex.2011/10/10 15:49				It is transmitted in the form of the character strings whose language code was selected by the programming pendant and half- and full-width characters are mixed.
5					
6					
7					
8	Alarm character strings name (character strings: 32 letters)				
9					
10					
11					
12					
13					
14					
15					



For the alarm character strings name, it is transmitted in the form of the character strings whose language code was selected by the programming pendant.

Use the same language code as the DX100, or the characters corrupt in case the client side dose not correspond to its language code.

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.2 Alarm History Reading Command

Request

Sub header part

<Details>

Command No.	0x71
Instance	Specify one out of followings • 1 to 100 • 1001 to 1100 • 2001 to 2100 • 3001 to 3100 • 4001 to 4100
Attribute	Specify one out of followings 1: Alarm code 2: Alarm data 3: Alarm type 4: Alarm occurring time 5: Alarm character strings name
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01

Specify the alarm number
 1 to 100 : Major failure
 1001 to 1100: Monitor alarm
 2001 to 2100: User alarm (system)
 3001 to 3100: User alarm (user)
 4001 to 4100: OFF line alarm

Alarm code means the alarm No.
 Alarm data means the sub code which supports the alarm content. There are some cases that the sub code for the occurring alarm would not appear.

Specify the accessing method to the data.
 0x0E: Read out data of the specified element number
 0x01: Read out data of all the element number
 (In this case, specify 0 to the element number.)

Data part

No data part

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD
Added status	Error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

32bit Integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Alarm code				Range is from 0x0001 to 0x270F(decimal value: 9999)
2	Alarm data				Setting values vary in accordance with the contents of the alarm type. Also, some alarm are not displayed with the sub code. In this case, the value is 0 :0x0).
3	Alarm type				0 : No alarm 1 : Decimal UNSIGNED SHORT type (display example: [1]) 2 : UNSIGNED CHAR bit pattern (display example: [0000_0001]) 3 : User axis type (display example: [SLURBT]) 4 : Spacial coordinate type (display example: [XYZ]) 5 : Robot coordinate type (display example: [XYZRxRyRz]) 6 : Conveyor characteristic file (display example: [123]) 8 : Control group type (display example: [R1R2S1S2]) robot & station 9 : Decimal SHORT type (display example: [-1]) 10 : UNSIGNED SHORT bit pattern (display example: [0000_0000_0000_0001]) 11 : Control group type (display example: [R1]) for robot only 12 : Control group type (display example:[R1S1B1]) for robot, station and base 20 : Control group LOW/HIGH logical axis (display example: [R1: LOW SLURBT, HIGH SLURBT]) 21 : Control group MIN/MAX logical axis (display example: [R1: MIN SLURBT, MAX SLURBT]) 22 : Control group MIN/MAX spacial coordinate (display example: [R1: MIN XYZ, MAX XYZ]) 23 : Logical axis of both control group 1 and control group 2 (display example: [R1: SLURBT, R2: SLURBT]) 24 : Logical axis 1 and 2 of the control group (display example: [R1:SLURBT, SLURBT]) 25 : Logical axis of the control group and UNSIGNED CHAR type (display example: [R1: SLURBT, 1]) 27 : Control group and UNSIGNED CHAR type (display example: [R: 1])
4	Alarm occurring time (Character strings of 16 letters) Ex.2011/10/10 15:49				It is transmitted in the form of the character strings whose language code was selected by the programming pendant and half- and full-width characters are mixed.
5					
6					
7					
8	Alarm character strings name (character strings: 32 letters)				
9					
10					
11					
12					
13					
14					
15					



For the alarm character strings name, it is transmitted in the form of the character strings whose language code was selected by the programming pendant.

Use the same language code as the DX100, or the characters corrupt in case the client side dose not correspond to its language code.

3.3.3 Status Information Reading Command

Request

Sub header part

<Details>

Command No.	0x72
Instance	Fixed to "1".
Attribute	Specify one out of followings 1: Data 1 2: Data 2
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01

Specify "1".

Specify the status data number.
For the details of Data1 and Data 2, refer to "Details of data".

Specify the accessing method to the data.
0x0E: Read out data of the specified element number
0x01: Read out data of all the element number
(In this case, specify 0 to the element number.)

Data part

No data part

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD
Added status	

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Data 1			
2	Data 2			

<Details>

Refer to "Details of data".

Refer to "Details of data".

Details of data

Data 1	bit0	Step	Data 2	bit0	
	bit1	1 cycle		bit1	In hold status (by programming pendant)
	bit2	Automatic and continuous		bit2	In hold status (externally)
	bit3	Running		bit3	In hold status (by command)
	bit4	In-guard safe operation		bit4	Alarming
	bit5	Teach		bit5	Error occurring
	bit6	Play		bit6	Servo ON
	bit7	Command remote		bit7	

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.4 Executing Job Information Reading Command

Request

Sub header part

<Details>

Command No.	0x73
Instance	Specify one out of followings 1: Master task 2: Sub task 1 3: Sub task 2 4: Sub task 3 5: Sub task 4 6: Sub task 5 7: Sub task 6 8: Sub task 7 9: Sub task 8 10: Sub task 9 11: Sub task 10 12: Sub task 11 13: Sub task 12 14: Sub task 13 15: Sub task 14 16: Sub task 15
Attribute	Specify one out of followings 1: Job name 2: Line number 3: Step number 4: Speed override value
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01

Specify the status data number of the executing job information.

Specify the accessing method to the data.
0x0E: Read out data of the specified element number
0x01: Read out data of all the element number
(In this case, specify 0 to the element number)

Data part

No data part

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 :respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Job name (character strings: 32 letters)				Job name
2					Half-width character: 32 characters
3					Full-width character: 16 characters
4					
5					
6					
7					
8					
9	Line No. (0 to 9999)				Job line number
10	Step No. (1 to 9998)				Job step number
11	Speed override value				Speed override value



For the alarm character strings name, it is transmitted in the form of the character strings whose language code was selected by the programming pendant.

Use the same language code as the DX100, or the characters corrupt in case the client side dose not correspond to its language code.

Sub header part

Command No.	0x74
Instance	Specify one out of followings • 1 to 8 • 11 to 18 • 21 to 44 • 101 to 108 • 111 to 118
Attribute	Specify one out of followings 1: "Axis name" of the first axis 2: "Axis name" of the second axis 3: "Axis name" of the third axis 4: "Axis name" of the fourth axis 5: "Axis name" of the fifth axis 6: "Axis name" of the sixth axis 7: "Axis name" of the seventh axis 8: "Axis name" of the eighth axis
Service	•Get_Attribute_Single:0x0E •Get_Attribute_All: 0x01

1	: R1 to 8	: R8	...Robot (pulse value)
11	: B1 to 18	: B8	...Base (pulse value)
21	: S1 to 44	: S24	...Station (pulse value)
101	: R1 to 108	: R8	...Robot (cartesian coordinate)
111	: B1 to 118	: B8	...Base (cartesian coordinate)

Specify the data number of axis information.
Each axis is justified for setting.
“0” is set to nonexistent axis.

Specify the accessing method to the data.

0x0E: Read out data of the specified element number.

0x01: Read out data of all the element number.
(In this case, specify 0 to the element number.)

No data part

Sub header part

Status	Respond by one in the followings • 0x00 :respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	First coordinate name				"S" (R*: pulse)/"X" (R*/B*: cartesian value)/ "1" (B*/S*: pulse)
2	Second coordinate name				"L" (R*: pulse)/"Y" (R*/B*: cartesian value)/ "2" (B*/S*: pulse)
3	Third coordinate name				"U" (R*: pulse)/"Z" (R*/B*: cartesian value)/ "3" (B*/S*: pulse)
4	Fourth coordinate name				"R" (R*: pulse)/"Rx" (R*: cartesian value)/ "4" (B*/S*: pulse)
5	Fifth coordinate name				"B" (R*: pulse)/"Ry" (R*: cartesian value)/ "5" (B*/S*: pulse)
6	Sixth coordinate name				"T" (R*: pulse)/"Rz" (R*: cartesian value)/ "6" (B*/S*: pulse)
7	Seventh coordinate name				"E" (R*: pulse)/"Rz" (R*: cartesian value)/ "7" (B*/S*: pulse)
8	Eighth coordinate name				

*: Each control group number.

R: Robot (R1 to R8)

S: Station (S1 to s24)

B: Base (B1 to b8)

Cartesian value can select the base coordinate only. (It cannot select the robot, user and tool coordinates.)

Sub header part

Command No.	0x75	
Instance	Specify one out of followings • 1 to 8 • 11 to 18 • 21 to 44 • 101 to 108	Specify the control group 1 : R1 to 8 : R8 ... Robot (pulse value) 11 : B1 to 18 : B8 ... Base (pulse value) 21 : S1 to 44 : S24 ... Station (pulse value) 101 : R1 to 108 : R8 ... Robot (cartesian coordinate)
Attribute	Specify one out of followings 1: Data type 2: Form 3: Tool number 4: User coordinate number 5: Extended form 6: First axis data 7: Second axis data 8: Third axis data 9: Fourth axis data 10: Fifth axis data 11: Sixth axis data 12: Seventh axis data 13: Eighth axis data	Specify the position information data number. 1 0: pulse value/16: base coordinate value 2 As for the form, refer to the “Details of data”. 3 Tool number 4 User coordinate number 5 As for the extended form, refer to the “Details of data”. 6 First axis data 7 Second axis data 8 Third axis data 9 Fourth axis data 10 Fifth axis data 11 Sixth axis data 12 Seventh axis data 13 Eighth axis data Each axis data is output by the same sequence as mentioned in <i>section 3.3.5 “Axis Configuration Information Reading Command”</i> , and “0” is set to nonexistent axis.
Service	•Get_Attribute_Single: 0x0E •Get_Attribute_All: 0x01	Specify the accessing method to the data. 0x0E: Read out data of the specified element number 0x01: Read out data of all the element number (In this case, specify 0 to the element number.)

Data part

No data part

Detail of data

Please refer section 3.9.5 “Flip/ No flip” in the “DX100 OPERATOR’S MANUAL” prepared for each application.

Form	bit0	0: Front	1: Back	Extended form	bit0	0: $\theta L < 180$,	1: $\theta L \geq 180$
	bit1	0: Upper arm	1: Lower arm		bit1	0: $\theta U < 180$,	1: $\theta U \geq 180$
	bit2	0: Flip	1: No flip		bit2	0: $\theta B < 180$,	1: $\theta B \geq 180$
	bit3	0: $\theta R < 180$,	1: $\theta R \geq 180$		bit3	0: $\theta E < 180$,	1: $\theta E \geq 180$
	bit4	0: $\theta T < 180$,	1: $\theta T \geq 180$		bit4	0: $\theta W < 180$,	1: $\theta W \geq 180$
	bit5	0: $\theta S < 180$,	1: $\theta S \geq 180$		bit5	Reserve	
	bit6	0: Redundant front	1: Redundant back		bit6	Reserve	
	bit7	0: Previous step regarded reverse conversion specified 1: Form regarded reverse conversion specified			bit7	Reserve	

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Data type				0: Pulse value/ 16: Base coordinate value
2	Form				For the form, refer to "Details of data".
3	Tool number				Tool number
4	User coordinate number				User coordinate number
5	Extended form				For the extended form, refer to "Details of data".
6	First axis data				
7	Second axis data				
8	Third axis data				
9	Fourth axis data				
10	Fifth axis data				
11	Sixth axis data				
12	Seventh axis data				
13	Eighth axis data				

Details of data

Please refer "3.9.5 Flip/ No flip" in "DX100 OPERATOR'S MANUAL" prepared for each application.

Form	bit0	0: Front	1: Back	Extended form	bit0	0: $\theta_L < 180$,	1: $\theta_L \geq 180$
	bit1	0: Upper arm	1: Lower arm		bit1	0: $\theta_U < 180$,	1: $\theta_U \geq 180$
	bit2	0: Flip	1: No flip		bit2	0: $\theta_B < 180$,	1: $\theta_B \geq 180$
	bit3	0: $\theta_R < 180$,	1: $\theta_R \geq 180$		bit3	0: $\theta_E < 180$,	1: $\theta_E \geq 180$
	bit4	0: $\theta_T < 180$,	1: $\theta_T \geq 180$		bit4	0: $\theta_W < 180$,	1: $\theta_W \geq 180$
	bit5	0: $\theta_S < 180$,	1: $\theta_S \geq 180$		bit5	Reserve	
	bit6	0: Redundant front	1: Redundant back		bit6	Reserve	
	bit7	0: Previous step regarded reverse conversion specified 1: Form regarded reverse conversion specified			bit7	Reserve	

3.3.7 Position Error Reading Command

Request

Sub header part

<Details>

Command No.	0x76
Instance	Specify one out of followings • 1 to 8 • 11 to 18 • 21 to 44
Attribute	Specify one out of followings 1: First axis data 2: Second axis data 3: Third axis data 4: Fourth axis data 5: Fifth axis data 6: Sixth axis data 7: Seventh axis data 8: Eighth axis data
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01

Specify the control group

 1 : R1 to 8 : R8 ... Robot axis
 11 : B1 to 18 : B8 ... Base axis
 21 : S1 to 44 : S24 ... Station axis

Specify the axis number.

 Each axis data is output by the same sequence as mentioned in *section 3.3.5 "Axis Configuration Information Reading Command"*, and "0" is set to nonexistent axis.

Specify the accessing method to the data.

 0x0E: Read out data of the specified element number
 0x01: Read out data of all the element number
 (In this case, specify 0 to the element number.)

Data part

No data part

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	First axis data			
2	Second axis data			
3	Third axis data			
4	Fourth axis data			
5	Fifth axis data			
6	Sixth axis data			
7	Seventh axis data			
8	Eighth axis data			

<Details>

Position variable data of each axis can be read out.

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.8 Torque Data Reading Data

Request

Sub header part

<Details>

Command No.	0x77
Instance	Specify one out of followings • 1 to 8 • 11 to 18 • 21 to 44
Attribute	Specify one out of followings 1: First axis data 2: Second axis data 3: Third axis data 4: Fourth axis data 5: Fifth axis data 6: Sixth axis data 7: Seventh axis data 8: Eighth axis data
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01

Specify the control group

1 : R1 to 8 : R8 ... Robot axis
 11 : B1 to 18 : B8 ... Base axis
 21 : S1 to 44 : S24 ... Station axis

Specify the axis number.

Each axis data is output by the same sequence as mentioned in *section 3.3.5 "Axis Configuration Information Reading Command"*, and "0" is set to nonexistent axis.

Specify the accessing method to the data.

0x0E: Read out data of the specified element number
 0x01: Read out data of all the element number
 (In this case, specify 0 to the element number.)

Data part

No data part

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	First axis data			
2	Second axis data			
3	Third axis data			
4	Fourth axis data			
5	Fifth axis data			
6	Sixth axis data			
7	Seventh axis data			
8	Eighth axis data			

<Details>

Torque data of each axis can be read out.

3.3.9 I/O Data Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x78	
Instance	Specify one out of followings • 1 to 256 • 1001 to 1256 • 2001 to 2256 • 3001 to 3256 • 4001 to 4160 • 5001 to 5200 • 6001 to 6064 • 7001 to 7999 • 2501 to 2756 • 3501 to 3756 • 8001 to 8064 • 8201 to 8220	Specify logical number /10 • 1 to 256 : Robot user input • 1001 to 1256: Robot user output • 2001 to 2256: External input • 2501 to 2756: Network input • 3001 to 3256: External output • 3501 to 3756: Network output • 4001 to 4160: Robot system input • 5001 to 5200: Robot system output • 6001 to 6064: Interface panel input • 7001 to 7999: Auxiliary relay • 8001 to 8064: Robot control status signal • 8201 to 8220: Pseudo input
Attribute	Fixed to "1".	Specify "1".
Service	• Get_Attribute_Single: 0x0E • Set_Attribute_Single: 0x10	Specify the accessing method to the data. 0x0E: Read out of all I/O data is enabler 0x01: Only network input signal is writable.

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	IO data				

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	IO data				

I/O data exists only when requested by the client.

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.10 Register Data Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x79	
Instance	Specify one out of followings • 0 to 999	Specify the register number 0 to 999 (writable register: 0 to 559)
Attribute	Fixed to "1".	Specify "1".
Service	• Get_Attribute_Single: 0x0E • Set_Attribute_Single: 0x10	Specify the accessing method to the data. 0x0E: Read out the specified register data 0x01: Register 0 to 599 is writable

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Register data				

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Register data				Register data exists only when requested by the client.

3.3.11 Byte Variable (B) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x7A	
Instance	Specify one out of followings • 0 to 99 (for standard setting)	Specify the variable number. Since the extended variable is an optional function, follow the numbers of the variables specified by the parameter when specifying the number.
Attribute	Fixed to "1".	Specify "1".
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01 • Set_Attribute_Single: 0x10 • Set_Attribute_All: 0x02	Specify the accessing method to the data. 0x0E/0x01: Read out data of the specified element number 0x10/0x02: Write the data to the specified variable

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	B variable				Set the data when writing.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	B variable				The data exists only when requested by the client.

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.12 Integer Type Variable (I) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x7B
Instance	Specify one out of followings • 0 to 99 (for standard setting)
Attribute	Fixed to "1".
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01 • Set_Attribute_Single: 0x10 • Set_Attribute_All: 0x02

Specify the variable number.
Since the extended variable is an optional function, follow the numbers of the variables specified by the parameter when specifying the number.

Specify "1".

Specify the accessing method to the data.
0x0E/0x01: Read out data of the specified element number
0x10/0x02: Write the data to the specified variable

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	I variable			

<Details>

Set the data when writing.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	I variable			

<Details>

The data exists only when requested by the client.

3.3.13 Double Precision Integer Type Variable (B) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x7C	
Instance	Specify one out of followings • 0 to 99 (for standard setting)	Specify the variable number. Since the extended variable is an optional function, follow the numbers of the variables specified by the parameter when specifying the number.
Attribute	Fixed to "1".	Specify "1".
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01 • Set_Attribute_Single: 0x10 • Set_Attribute_All: 0x02	Specify the accessing method to the data. 0x0E/0x01: Read out data of the specified element number 0x10/0x02: Write the data to the specified variable

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	D variable				Set the data when writing.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	D variable				The data exists only when requested by the client.

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.14 Real Type Variable (R) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x7D
Instance	Specify one out of followings • 0 to 99 (for standard setting)
Attribute	Fixed to "1".
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01 • Set_Attribute_Single: 0x10 • Set_Attribute_All: 0x02

Specify the variable number.
Since the extended variable is an optional function, follow the numbers of the variables specified by the parameter when specifying the number.

Specify "1".

Specify the accessing method to the data.
0x0E/0x01: Read out data of the specified element number
0x10/0x02: Write the data to the specified variable

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	R variable			

<Details>

Set the data when writing.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	R variable			

<Details>

The data exists only when requested by the client.

3.3.15 Character Type Variable (S) Reading Writing Command

Request

Sub header part

<Details>

Command No.	0x7E	
Instance	Specify one out of followings • 0 to 99 (for standard setting)	Specify the variable number. Since the extended variable is an optional function, follow the numbers of the variables specified by the parameter when specifying the number
Attribute	Fixed to "1".	Specify "1".
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01 • Set_Attribute_Single: 0x10 • Set_Attribute_Al: 0x02	Specify the accessing method to the data. 0x0E/0x01: Read out data of the specified element number 0x10/0x02: Write the data to the specified variable

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	S variable				Set the data when writing.
2					
3					
4					

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	S variable				The data exists only when requested by the client.
2					
3					
4					

3.3.16 Robot Position Type Variable (P) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x7F	
Instance	Specify one out of followings • 0 to 127 (for standard setting)	Specify the variable number. Since the extended variable is an optional function, follow the numbers of the variables specified by the parameter when specifying the number.
Attribute	Specify one out of followings 1: Data type 2: Form 3: Tool number 4: User coordinate number 5: Extended form 6: "Coordinated data" of the first axis 7: "Coordinated data" of the second axis 8: "Coordinated data" of the third axis 9: "Coordinated data" of the fourth axis 10: "Coordinated data" of the fifth axis 11: "Coordinated data" of the sixth axis 12: "Coordinated data" of the seventh axis 13: "Coordinated data" of the eighth axis	Specify the axis information data number. Followings are the data type. 0: Pulse value 16: Base coordinated value 17: Robot coordinated value 18: User coordinated value 19: Tool coordinated value
Service	• Get_Attribute_All: 0x01 • Set_Attribute_All: 0x02	Specify the accessing method to the data. 0x0E/0x01: Read out data of the specified element number 0x10/0x02: Write the data to the specified variable

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Data type				0: Pulse value 16: Base coordinated value 17: Robot coordinated value 18: User coordinated value 19: Tool coordinated value
2	Form				For the form, refer to "Details of data".
3	Tool number				Tool number
4	User coordinate number				User coordinate number
5	Extended form				For the extended form, refer to "Details of data".
6	First coordinate data				
7	Second coordinate data				
8	Third coordinated data				
9	Fourth coordinate data				
10	Fifth coordinate data				
11	Sixth coordinate data				
12	Seventh coordinate data				
13	Eighth coordinate data				

Details of data

Please refer “3.9.5 Flip/ No flip” in “DX100 OPERATOR’S MANUAL” prepared for each application.

Form	bit0	0: Front	1: Back	Extended form	bit0	0: $\theta L < 180$,	1: $\theta L \geq 180$
	bit1	0: Upper arm	1: Lower arm		bit1	0: $\theta U < 180$,	1: $\theta U \geq 180$
	bit2	0: Flip	1: No flip		bit2	0: $\theta B < 180$,	1: $\theta B \geq 180$
	bit3	0: $\theta R < 180$,	1: $\theta R \geq 180$		bit3	0: $\theta E < 180$,	1: $\theta E \geq 180$
	bit4	0: $\theta T < 180$,	1: $\theta T \geq 180$		bit4	0: $\theta W < 180$,	1: $\theta W \geq 180$
	bit5	0: $\theta S < 180$,	1: $\theta S \geq 180$		bit5	Reserve	
	bit6	0: Redundant front	1: Redundant back		bit6	Reserve	
	bit7	0: Previous step regarded reverse conversion specified 1: Form regarded reverse conversion specified			bit7	Reserve	

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Data type				0: Pulse value 16: Base coordinated value 17: Robot coordinated value 18: User coordinated value 19: Tool coordinated value
2	Form				For the form, refer to "Details of data".
3	Tool number				Tool number
4	User coordinate number				User coordinate number
5	Extended form				For the extended form, refer to "Details of data".
6	First coordinate data				
7	Second coordinate data				
8	Third coordinated data				
9	Fourth coordinate data				
10	Fifth coordinate data				
11	Sixth coordinate data				
12	Seventh coordinate data				
13	Eighth coordinate data				

Details of data

Please refer to section 3.9.5 "Flip/ No flip" in "DX100 OPERATOR'S MANUAL" prepared for each application.

Form	bit0	0: Front	1: Back	Extended form	bit0	0: $\theta L < 180$,	1: $\theta L \geq 180$
	bit1	0: Upper arm	1: Lower arm		bit1	0: $\theta U < 180$,	1: $\theta U \geq 180$
	bit2	0: Flip	1: No flip		bit2	0: $\theta B < 180$,	1: $\theta B \geq 180$
	bit3	0: $\theta R < 180$,	1: $\theta R \geq 180$		bit3	0: $\theta E < 180$,	1: $\theta E \geq 180$
	bit4	0: $\theta T < 180$,	1: $\theta T \geq 180$		bit4	0: $\theta W < 180$,	1: $\theta W \geq 180$
	bit5	0: $\theta S < 180$,	1: $\theta S \geq 180$		bit5	Reserve	
	bit6	0: Redundant front	1: Redundant back		bit6	Reserve	
	bit7	0: Previous step regarded reverse conversion specified 1: Form regarded reverse conversion specified			bit7	Reserve	

3.3.17 Base Position Type Variable (Bp) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x80	
Instance	Specify one out of followings • 0 to 127 (for standard setting)	Specify the variable number. Since the extended variable is an optional function, follow the numbers of the variables specified by the parameter when specifying the number.
Attribute	Specify one out of followings 1: Data type 2: "Coordinated data" of the first axis 3: "Coordinated data" of the second axis 4: "Coordinated data" of the third axis 5: "Coordinated data" of the fourth axis 6: "Coordinated data" of the fifth axis 7: "Coordinated data" of the sixth axis 8: "Coordinated data" of the seventh axis 9: "Coordinated data" of the eighth axis	Specify the axis information data number. Followings are the data type. 0: Pulse value 16: Base coordinated value
Service	• Get_Attribute_Single :0x0E • Get_Attribute_All :0x01 • Set_Attribute_Single :0x10 • Set_Attribute_All :0x02	Specify the accessing method to the data. 0x0E: Read out the specified data 0x01: Read out the data 0x10: Write a specified data. If it is not an object element, keep the data previous to writing operation. 0x02: Write the data

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Data type				0: Pulse value 16: Base coordinated value
2	First coordinate data				
3	Second coordinate data				
4	Third coordinated data				
5	Fourth coordinate data				
6	Fifth coordinate data				
7	Sixth coordinate data				
8	Seventh coordinate data				
9	Eighth coordinate data				

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Data type			
2	First coordinate data			
3	Second coordinate data			
4	Third coordinated data			
5	Fourth coordinate data			
6	Fifth coordinate data			
7	Sixth coordinate data			
8	Seventh coordinate data			
9	Eighth coordinate data			

<Details>

0: Pulse value

16: Base coordinated value

3.3.18 External Axis Type Variable (Ex) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x81	
Instance	Specify one out of followings • 0 to 127 (for standard setting)	Specify the variable number. Since the extended variable is an optional function, follow the numbers of the variables specified by the parameter when specifying the number.
Attribute	Specify one out of followings 1: Data type 2: "Coordinated data" of the first axis 3: "Coordinated data" of the second axis 4: "Coordinated data" of the third axis 5: "Coordinated data" of the fourth axis 6: "Coordinated data" of the fifth axis 7: "Coordinated data" of the sixth axis 8: "Coordinated data" of the seventh axis 9: "Coordinated data" of the eighth axis	Specify the axis information data number. Followings are the data type. 0: Pulse value
Service	• Get_Attribute_Single :0x0E • Get_Attribute_All :0x01 • Set_Attribute_Single :0x10 • Set_Attribute_All :0x02	Specify the accessing method to the data. 0x0E : Read out the specified data 0x01 : Read out the data 0x10 :Write a specified data. If it is not an object element, keep the data previous to writing operation. 0x02 : Write the data

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Data type				0: Pulse value
2	First coordinate data				
3	Second coordinate data				
4	Third coordinated data				
5	Fourth coordinate data				
6	Fifth coordinate data				
7	Sixth coordinate data				
8	Seventh coordinate data				
9	Eighth coordinate data				

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Data type			
2	First coordinate data			
3	Second coordinate data			
4	Third coordinated data			
5	Fourth coordinate data			
6	Fifth coordinate data			
7	Sixth coordinate data			
8	Seventh coordinate data			
9	Eighth coordinate data			

<Details>

0: Pulse value

3.3.19 Alarm Reset / Error Cancel Command

Request

Sub header part

<Details>

Command No.	0x82	
Instance	Specify one out of followings 1: Resetting of alarm 2: Cancelling of error	Specify the type of reset/cancel 1: RESET (resetting of alarm) 2: CANCEL (cancelling of error)
Attribute	Fixed to "1".	Specify "1".
Service	• Set_Attribute_Single: 0x10	Specify the accessing method to the data. 0x10 : Execute the specified request

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Data 1				Fixed to "1".

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

No data part

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.20 Hold / Servo On/off Command

Request

Sub header part

<Details>

Command No.	0x83	
Instance	Specify one out of followings 1: HOLD 2: Servo ON 3: HLOCK	Specify the type of OFF/ON command 1: HOLD 2: Servo ON 3: HLOCK (Refer to "Details of data".)
Attribute	Fixed to "1".	Specify "1".
Service	• Set_Attribute_Single: 0x10	Specify the accessing method to the data. 0x10 : Execute the specified request

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	1:ON 2:OFF				Specify ON/OFF

Details of data

■ HLOCK

This data interlocks the P.P and I/O operation system signals. Only the following operations are available while the interlock operation is ON.

- Emergency stop for the programming pendant
- Inputting signals excluding I/O mode switching, external start, external servo ON, cycle switch, inhibit I/O, inhibit PP/PANEL and master calling up.

HLOCK is invalid while the programming pendant is in edit mode or it is file accessing using other functions.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

3.3.21 Step / Cycle / Continuous Switching Command

Request

Sub header part

<Details>

Command No.	0x84
Instance	Specify the following • 2
Attribute	Fixed to “1”.
Service	• Set_Attribute_Single: 0x10

Specify the type of status switch command
2: CYCLE (switching of STEP/CYCLE/CONTINUE)
Specify “1”.
Specify the accessing method to the data.
0x10 : Execute the specified request

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Data 1			

<Details>

CYCLE = 1: STEP/2: 1 CYCLE/3:CONTINUE

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.22 Character String Display Command To The Programming Pendant

Request

Sub header part

<Details>

Command No.	0x85
Instance	Fixed to "1".
Attribute	Fixed to "1".
Service	• Set_Attribute_Single: 0x10

Specify "1".

Specify "1".

Specify the accessing method to the data.
0x10 : Execute the specified request

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Displaying message			
2				
3				
4				
5				
6				
7				
8				

<Details>

Set the character strings to be indicated on the programming pendant
Half-width character: 30 characters
Full-width character: 15 characters

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

No data part



For the alarm character strings name, it is transmitted in the form of the character strings whose language code was selected by the programming pendant.

Use the same language code as the DX100, or the characters corrupt in case the client side dose not correspond to its language code.

3.3.23 Start-up (Job Start) Command

Request

Sub header part

<Details>

Command No.	0x86
Instance	Fixed to “1”.
Attribute	Fixed to “1”.
Service	• Set_Attribute_Single: 0x10

Specify “1”.
Specify “1”.
Specify the accessing method to the data.
0x10 : Execute the specified request

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Data 1			

<Details>
Fixed to “1”.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part

3.3.24 Job Select Command

Request

Sub header part

<Details>

Command No.	0x87	
Instance	Specify one out of followings 1: Set the executing job 10: Set the master job (task 0) 11: Set the master job (task 1) 12: Set the master job (task 2) 13: Set the master job (task 3) 14: Set the master job (task 4) 15: Set the master job (task 5) 16: Set the master job (task 6) 17: Set the master job (task 7) 18: Set the master job (task 8) 19: Set the master job (task 9) 20: Set the master job (task 10) 21: Set the master job (task 11) 22: Set the master job (task 12) 23: Set the master job (task 13) 24: Set the master job (task 14) 25: Set the master job (task 15)	Specify the type.
Attribute	Specify one out of followings 1: Job name 2: Line number (valid only when executing job setting.)	Specify the setting content.
Service	• Set_Attribute_All: 0x02	Specify the accessing method to the data. 0x02: Read out data of all the element number (In this case, specify 0 to the element number.)

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte 3	<Details>
1	Job name (Character strings: 32 characters)				Job name
2					Half-width character: 32 characters
3					Full-width character: 16 characters
4					
5					
6					
7					
8					
9	Line number (0 to 9999)				Line number



For the alarm character strings name, it is transmitted in the form of the character strings whose language code was selected by the programming pendant.

Use the same language code as the DX100, or the characters corrupt in case the client side dose not correspond to its language code.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.25 Management Time Acquiring Command

Request

Sub header part

<Details>

Command No.	0x88	
Instance	Specify one out of followings • 1 • 10 • 11 to 18 • 21 to 44 • 110 • 111 to 118 • 121 to 144 • 210 • 211 to 218 • 221 to 244 • 301 to 308	Specify the type of the management time 1 :Control power ON time 10 :Servo power ON time (TOTAL) 11 to 18 :Servo power ON time (R1 to R8) 21 to 44 :Servo power ON time (S1 to S24) 110 :Play back time (TOTAL) 111 to 118 :Play back time (R1 to R8) 121 to 144 :Play back time (S1 to S24) 210 :Motion time (TOTAL) 211 to 218 :Motion time (R1 to R8) 221 to 244 :Motion time (S1 to S24) 301 to 308 :Operation time (application 1 to 8)
Attribute	Specify one out of followings 1: Operation start time 2: Elapse time	Specify the type of the management time
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01	Specify the accessing method to the data. 0x0E : Read out data of the specified element number 0x01 : Read out data of all the element number (In this case, specify 0 to the element number.)

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Operation start time (Character strings: 16 characters) Ex. 2011/10/10 15:49				Operation start time
2					
3					
4					
5	Elapse time (Character strings: 12 characters) Ex. 000000:00'00				Elapse time
6					
7					

3.3.26 System Information Acquiring Command

Request

Sub header part

<Details>

Command No.	0x89	
Instance	Specify one out of followings • 11 to 18 • 21 to 44 • 101 to 108	Specify the type of system type. 11 to 18: Type information (R1 to R8) 21 to 44: Type information (S1 to S24) 101 to 108: Application information (application 1 to 8)
Attribute	Specify one out of followings 1: System software version 2: Model name / application 3: Parameter version	Specify the type of system information
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01	Specify the accessing method to the data. 0x0E: Read out data of the specified element number 0x01: Read out data of all the element number (In this case, specify 0 to the element number)

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	System software version (Character strings: 24 characters) Ex. DS2.07.00A. (JP/US) -00				The same character strings are returned even if either 11 to 18, 21 to 44 or 101 to 108 is specified to the instance in the request sub-header part.
2					
3					
4					
5					
6					
7	Model name / application (Character strings: 16 characters) Ex. (For model) ES0165D-A0* (For application) ARC WELDING				The model name is returned when it is R1 to R8, and NULL character is returned when it is S1 to S24. Also, application name is returned when it is application 1 to 8R.
8					
9					
10					
11	Parameter version (Character strings: 8 characters) Ex. 12.34				R1 to R8: Parameter version
12					When it is nonexistent control group, it is returned in NULL characters.

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.27 Plural I/o Data Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x300	
Instance	Specify one out of followings • 1 to 256 • 1001 to 1256 • 2001 to 2256 • 3001 to 3256 • 4001 to 4160 • 5001 to 5200 • 6001 to 6064 • 7001 to 7999 • 2501 to 2756 • 3501 to 3756 • 8001 to 8064 • 8201 to 8220	Specify logical number /10 • 1 to 256 : Robot user input • 1001 to 1256: Robot user output • 2001 to 2256: External input • 2501 to 2756: Network input • 3001 to 3256: External output • 3501 to 3756: Network output • 4001 to 4160: Robot system input • 5001 to 5200: Robot system output • 6001 to 6064: Interface panel input • 7001 to 7999: Auxiliary relay • 8001 to 8064: Robot control status signal • 8201 to 8220: Pseudo input
Attribute	Fixed to "0".	Specify "0".
Service	0x33:Read plural data 0x34:Write plural data	Specify the accessing method to the data. 0x33: Read out the fixed size specified by the data part. 0x34: Write the fixed size specified by the data part. Only the network input signal can be writable.

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 474 *It can specify by a multiple of 2 only.
2	I/O data 1	I/O data 2	I/O data 3	I/O data 4	I/O data part is valid only when writing. Only the number of data is valid when reading.
:					
120	I/O data 473	I/O data 474			

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part
(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details> Maximum: 474 *It can specify by a multiple of 2 only. I/O data part is valid only when writing. Only the number of data is valid when reading.
1	Number				
2	I/O data 1	I/O data 2	I/O data 3	I/O data 4	
:					
120	I/O data 473	I/O data 474			

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.28 Plural Register Data Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x301	
Instance	Specify one out of followings • 0 to 999	Specify the variable number (the first number with which reading/writing is executed) 0 to 999 (writable register: 0 to 559)
Attribute	Fixed to "0"	Specify "0"
Service	0x33 : Read plural data 0x34 : Write plural data	Specify the accessing method to the data. 0x33: Read out the fixed size specified by the data part. 0x34: Write the fixed size specified by the data part.

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 237
2	Register data 1		Register data 2		I/O data part is valid only when writing. Only the number of data is valid when reading.
:					
120	Register data 237				

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 237
2	Register data 1		Register data 2		The data part is valid only when requested by the client.
:					
120	Register data 237				

3.3.29 Plural Byte Type Variable (B) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x302
Instance	Specify one out of followings • 0 to 99 (for standard setting)
Attribute	Fixed to "0".
Service	0x33 : Read plural data 0x34 : Write plural data

Specify the variable number (the first number with which reading/writing is executed)
Follow the numbers of the variable specified by the parameter since the extended variable is an optional function.

Specify "0".

Specify the accessing method to the data.
0x33: Read out the fixed size specified by the data part.
0x34: Write the fixed size specified by the data part.

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2	B variable 1	B variable 2	B variable 3	B variable 4
	:			
120	B variable 473	B variable 474		

<Details>

Maximum: 474
*It can specify by a multiple of 2 only.

Variable data part is valid only when writing. Only the number of data is valid when reading.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

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3.3 Respective Commands for Robot Control

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2	B variable 1	B variable 2	B variable 3	B variable 4

<Details>

Maximum: 474
*It can specify by a multiple of 2 only. (invalid if specified by other than a multiple of 2)

:

120	B variable 473	B variable 474
-----	----------------	----------------

3.3.30 Plural Integer Type Variable (I) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x303
Instance	Specify one out of followings • 0 to 99 (for standard setting)
Attribute	Fixed to "0"
Service	0x33 : Read plural data 0x34 : Write plural data

Specify the variable number (the first number with which reading/writing is executed)
Follow the numbers of the variable specified by the parameter since the extended variable is an optional function.

Specify "0"
Only batch access of all elements is valid

Specify the accessing method to the data.
0x33: Read plural data.
0x34: Write plural data

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2	I variable 1		I variable 2	
:				
120	I variable 237			

<Details>

Maximum: 237

Variable data part is valid only when writing. Only the number of data is valid when reading.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2	I variable 1		I variable 2	
:				
120	I variable 237			

<Details>

Maximum: 237

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.31 Plural Double Precision Integer Type Variable (B) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x304	
Instance	Specify one out of followings • 0 to 99 (for standard setting)	Specify the variable number (the first number with which reading/writing is executed) Follow the numbers of the variable specified by the parameter since the extended variable is an optional function.
Attribute	Fixed to "0"	Specify "0" Only batch access of all elements is valid
Service	0x33 : Read plural data 0x34 : Write plural data	Specify the accessing method to the data. 0x33: Read plural data 0x34: Write plural data

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 118
2	D variable 1				Variable data part is valid only when writing. Only the number of data is valid when reading.
:					
119	D variable 118				

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 118
2	D variable 1				
:					
119	D variable 118				

3.3.32 Plural Real Type Variable (R) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x305
Instance	Specify one out of followings • 0 to 99 (for standard setting)
Attribute	Fixed to "0"
Service	0x33 : Read plural data 0x34 : Write plural data

Specify the variable number (the first number with which reading/writing is executed)
Follow the numbers of the variable specified by the parameter since the extended variable is an optional function.

Specify "0"
Only batch access of all elements is valid

Specify the accessing method to the data.
0x33: Read plural data
0x34: Write plural data

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2	R variable 1			
	:			
119	R variable 118			

<Details>

Maximum: 118

Variable data part is valid only when writing. Only the number of data is valid when reading.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2	R variable 1			
	:			
119	R variable 118			

<Details>

Maximum: 118

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.33 Plural Character Type Variable (S) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x306	
Instance	Specify one out of followings • 0 to 99 (for standard setting)	Specify the variable number (the first number with which reading/writing is executed) Follow the numbers of the variable specified by the parameter since the extended variable is an optional function.
Attribute	Fixed to "0"	Specify "0" Only batch access of all elements is valid
Service	0x33 : Read plural data 0x34 : Write plural data	Specify the accessing method to the data. 0x33: Read plural data 0x34: Write plural data

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 29
2	S variable 1				Variable data part is valid only when writing.
3					Only the number of data is valid when
4					reading.
5					
					:
114	S variable 29				
115					
116					
117					

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	The error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part
(Data exists during the writing operation only)

32bit integer		Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number					Maximum: 29
2	S variable 1					
3						
4						
5						
:						
114	S variable 29					
115						
116						
117						

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.34 Plural Robot Position Type Variable (P) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x307	
Instance	Specify one out of followings • 0 to 127 (for standard setting)	Specify the variable number (the first number with which reading/writing is executed) Follow the numbers of the variable specified by the parameter since the extended variable is an optional function.
Attribute	Fixed to "0"	Specify "0" Only batch access of all elements is valid
Service	0x33 : Read plural data 0x34 : Write plural data	Specify the accessing method to the data. 0x33: Read plural data 0x34: Write plural data

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 9
2 to 14	Data type				0: Pulse value 16: Base coordinated value 17: Robot coordinated value 18: User coordinated value 19: Tool coordinated value
	Form				Form
	Tool number				Tool number
	User coordinate number				User coordinate number
	Extended form				
	First coordinate data				
	Second coordinate data				
	Third coordinated data				
	Fourth coordinate data				
	Fifth coordinate data				Variable data part is valid only when writing.
	Sixth coordinate data				Only the number of data is valid when reading.
	Seventh coordinate data				
	Eighth coordinate data				
:					
106 to 118	Data type				0: Pulse value 16: Base coordinated value 17: Robot coordinated value 18: User coordinated value 19: Tool coordinated value
	Form				Form
	Tool number				Tool number
	User coordinate number				User coordinate number
	Extended form				
	First coordinate data				
	Second coordinate data				
	Third coordinated data				
	Fourth coordinate data				
	Fifth coordinate data				
	Sixth coordinate data				
	Seventh coordinate data				
	Eighth coordinate data				

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2 to 14	Data type			
	Form			
	Tool number			
	User coordinate number			
	Extended form			
	First coordinate data			
	Second coordinate data			
	Third coordinated data			
	Fourth coordinate data			
	Fifth coordinate data			
	Sixth coordinate data			
	Seventh coordinate data			
	Eighth coordinate data			

<Details>

Maximum: 9

0: Pulse value
16: Base coordinated value
17: Robot coordinated value
18: User coordinated value
19: Tool coordinated value

Form

Tool number

User coordinate number

Variable data part is valid only when writing.
Only the number of data is valid when reading.

:

106 to 118	Data type			
	Form			
	Tool number			
	User coordinate number			
	Extended form			
	First coordinate data			
	Second coordinate data			
	Third coordinated data			
	Fourth coordinate data			
	Fifth coordinate data			
	Sixth coordinate data			
	Seventh coordinate data			
	Eighth coordinate data			

0: Pulse value
16: Base coordinated value
17: Robot coordinated value
18: User coordinated value
19: Tool coordinated value

Form

Tool number

User coordinate number

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.35 Plural Base Position Type Variable (Bp) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x308	
Instance	Specify one out of followings • 0 to 127 (for standard setting)	Specify the variable number (the first number with which reading/writing is executed) Follow the numbers of the variable specified by the parameter since the extended variable is an optional function.
Attribute	Fixed to "0".	Specify "0".
Service	0x33 : Read plural data 0x34 : Write plural data	Specify the accessing method to the data. 0x33: Read plural data 0x34: Write plural data

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 13
2 (Replying data is determined by the value specified by the element number.)	Data type				0x00 : Pulse value 0x10 : Base coordinate value
	First coordinate data				
	Second coordinate data				
	Third coordinated data				
	Fourth coordinate data				
	Fifth coordinate data				Variable data part is valid only when writing.
	Sixth coordinate data				Only the number of data is valid when reading.
	Seventh coordinate data				
	Eighth coordinate data				
:					
119	Data type				0x00 : Pulse value 0x10 : Base coordinate value
	First coordinate data				
	Second coordinate data				
	Third coordinated data				
	Fourth coordinate data				
	Fifth coordinate data				
	Sixth coordinate data				
	Seventh coordinate data				
	Eighth coordinate data				

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2 to 10 (Replying data is determined by the value specified by the element number.)	Data type			
	First coordinate data			
	Second coordinate data			
	Third coordinated data			
	Fourth coordinate data			
	Fifth coordinate data			
	Sixth coordinate data			
	Seventh coordinate data			
	Eighth coordinate data			

<Details>

Maximum: 13

0x00 : Pulse value

0x10 : Base coordinate

Variable data part is valid only when writing.
Only the number of data is valid when reading.

119	Data type			
	First coordinate data			
	Second coordinate data			
	Third coordinated data			
	Fourth coordinate data			
	Fifth coordinate data			
	Sixth coordinate data			
	Seventh coordinate data			
	Eighth coordinate data			

0x00 : Pulse value

0x10 : Base coordinate

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.36 Plural External Axis Type Variable (Ex) Reading / Writing Command

Request

Sub header part

<Details>

Command No.	0x309	
Instance	Specify one out of followings • 0 to 127 (for standard setting)	Specify the variable number (the first number with which reading/writing is executed) Follow the numbers of the variable specified by the parameter since the extended variable is an optional function.
Attribute	Fixed to "0"	Specify "0".
Service	0x33 : Read plural data 0x34 : Write plural data	Specify the accessing method to the data. 0x33: Read plural data 0x34: Write plural data

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Number				Maximum: 13
2 to 10	Data type				0 : Pulse value
	First coordinate data				
	Second coordinate data				
	Third coordinated data				
	Fourth coordinate data				
	Fifth coordinate data				Variable data part is valid only when writing.
	Sixth coordinate data				Only the number of data is valid when
	Seventh coordinate data				reading.
	Eighth coordinate data				
:					
110 to 118	Data type				0 : Pulse value
	First coordinate data				
	Second coordinate data				
	Third coordinated data				
	Fourth coordinate data				
	Fifth coordinate data				
	Sixth coordinate data				
	Seventh coordinate data				
	Eighth coordinate data				

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

(Data exists during the writing operation only)

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
1	Number			
2 to 10	Data type			
	First coordinate data			
	Second coordinate data			
	Third coordinated data			
	Fourth coordinate data			
	Fifth coordinate data			
	Sixth coordinate data			
	Seventh coordinate data			
	Eighth coordinate data			

<Details>

Maximum: 13

0: Pulse value

Variable data part is valid only when writing.
Only the number of data is valid when reading.

:

110 to 118	Data type			
	First coordinate data			
	Second coordinate data			
	Third coordinated data			
	Fourth coordinate data			
	Fifth coordinate data			
	Sixth coordinate data			
	Seventh coordinate data			
	Eighth coordinate data			

0: Pulse value

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3 Transmission Procedure

3.3 Respective Commands for Robot Control

3.3.37 Alarm Data Reading Command (for Applying the Sub Code Character Strings)

Request

Sub header part

<Details>

Command No.	0x30A	
Instance	Specify one out of followings 1: The latest alarm 2: The second alarm from the latest 3: The third alarm from the latest 4: The fourth alarm from the latest	Up to four alarms are displayed on the P.P display at the same time. Specify one out of them.
Attribute	Specify one out of followings 1: Alarm code 2: Alarm data 3: By alarm type 4: Alarm occurring time 5: Alarm character string name 6: Sub code data additional information character strings 7: Sub code data character strings 8: Sub code data character strings reverse display information	Alarm code means the alarm No. Alarm data means the sub code which supports the alarm contents. Some alarms may not appear as the sub code. Sub code additional info character strings means the number for alarms from the Servo circuit board [SV#*] or the function safety board [NSU(CPU#*)]. (*denotes number) Sub code data character string reverse display information sets [1], when the characters are reverse.
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01	Specify the accessing method to the data. 0x0E: Read out data of the specified element number 0x01: Read out data of all the element number (In this case, specify 0 to the element number.)

Data part

No data part

Answer

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally	
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD	"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.
Added status	Error code specified by the added status size	The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte 3	<Details>
1	Alarm code				Range is from 0x0001 to 0x270F(decimal value: 9999)
2	Alarm data				Setting values vary in accordance with the contents of the alarm type. Also, some alarms are not displayed with the sub code. In this case, the value is zero (0x0).
3	Alarm type				0 : No alarm 1 : Decimal UNSIGNED SHORT type (display example: [1]) 2 : UNSIGNED CHAR bit pattern (display example: [0000_0001]) 3 : User axis type (display example: [SLURBT]) 4 : Spacial coordinate type (display example: [XYZ]) 5 : Robot coordinate type (display example: [XYZRxRyRz]) 6 : Conveyor characteristic file (display example: [123]) 8 : Control group type (display example: [R1R2S1S2]) robot & station 9 : Decimal SHORT type (display example: [-1]) 10 : UNSIGNED SHORT bit pattern (display example: [0000_0000_0000_0001]) 11 : Control group type (display example: [R1]) for robot only 12 : Control group type (display example:[R1S1B1]) for robot, station and base 20 : Control group LOW/HIGH logical axis (display example: [R1:LOW SLURBT, HIGH SLURBT]) 21 : Control group MIN/MAX logical axis (display example: [R1: MIN SLURBT, MAX SLURBT]) 22 : Control group MIN/MAX spacial coordinate (display example: [R1: MIN XYZ, MAX XYZ]) 23 : Logical axis of both control group 1 and control group 2 (display example: [R1: SLURBT, R2: SLURBT]) 24 : Logical axis 1 and 2 of the control group (display example: [R1: SLURBT, SLURBT]) 25 : Logical axis of the control group and UNSIGNED CHAR type (display example: [R1: SLURBT, 1]) 27 : Control group and UNSIGNED CHAR type (display example: [R1: 1])
4 to 7	Alarm occurring time (Character strings of 16 letters) Ex.2011/10/10 15:49				
8 to 15	Alarm character strings name (character strings: 32 letters)				It is transmitted in the form of the character strings whose language code was selected by the programming pendant and half- and full-width characters are mixed.
16 to 19	Sub code data additional information character strings (Character strings of 16 letters)				[SV#1] indicates the servo board number 1. [NSU(CPU#1)] indicates that an alarm is found in the function safety unit CPU#1.
20 to 43	Sub code data character strings (Character strings of 96 letters)				
44 to 67	Sub code data character strings reverse display information (Character strings of 96 letters)				Regular characters show [0] and reverse characters show [1]. (display example: [R2S1S2])

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For the alarm character strings name, it is transmitted in the form of the character strings whose language code was selected by the programming pendant.

Use the same language code as the DX100, or the characters corrupt in case the client side dose not correspond to its language code.

3.3.38 Alarm History Reading Command (for Applying the Sub Code Character Strings)

Request

Sub header part

<Details>

Command No.	0x30B
Instance	Specify one out of followings • 1 to 100 • 1001 to 1100 • 2001 to 2100 • 3001 to 3100 • 4001 to 4100
Attribute	Specify one out of followings 1:Alarm code 2:Alarm data 3:Alarm type 4:Alarm occurring time 5:Alarm character strings name 6:Sub code data additional information character strings 7:Sub code data character strings 8:Sub code data character strings reverse display information
Service	• Get_Attribute_Single: 0x0E • Get_Attribute_All: 0x01

Specify the alarm number
 1 to 100 : Major failure
 1001 to 1100: Monitor alarm
 2001 to 2100: User alarm (system)
 3001 to 3100: User alarm (user)
 4001 to 4100: OFF line alarm

Alarm code means the alarm No.
 Alarm data means the sub code which supports the alarm content. Some alarms may not appear as the sub code.
 Sub code additional info character strings mean the number for alarms from the Servo circuit board [SV#*] or the function safety board [NSU(CPU#*)].
 Sub code data character strings reverse display information means setting [1], when the characters are reverse.

Specify the accessing method to the data.
 0x0E: Read out data of the specified element number
 0x01: Read out data of all the element number
 (In this case, specify 0 to the element number.)

Data part

No data part

Answer

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD
Added status	Error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

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3.3 Respective Commands for Robot Control

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Alarm code				Range is from 0x0001 to 0x270F(decimal value: 9999)
2	Alarm data				Setting values vary in accordance with the contents of the alarm type. Also, some alarms are not displayed with the sub code. In this case, the value is zero (0x0).
3	Alarm type				0 : No alarm 1 : Decimal UNSIGNED SHORT type (display example: [1]) 2 : UNSIGNED CHAR bit pattern (display example: [0000_0001]) 3 : User axis type (display example: [SLURBT]) 4 : Spacial coordinate type (display example: [XYZ]) 5 : Robot coordinate type (display example: [XYZRxRyRz]) 6 : Conveyor characteristic file (display example: [123]) 8 : Control group type (display example: [R1R2S1S2]) robot & station 9 : Decimal SHORT type (display example: [-1]) 10 : UNSIGNED SHORT bit pattern (display example: [0000_0000_0000_0001]) 11 : Control group type (display example: [R1]) for robot only 12 : Control group type (display example:[R1S1B1]) for robot, station and base 20 : Control group LOW/HIGH logical axis (display example: [R1:LOW SLURBT, HIGH SLURBT]) 21 : Control group MIN/MAX logical axis (display example: [R1: MIN SLURBT, MAX SLURBT]) 22 : Control group MIN/MAX spacial coordinate (display example: [R1: MIN XYZ, MAX XYZ]) 23 : Logical axis of both control group 1 and control group 2 (display example: [R1: SLURBT, R2: SLURBT]) 24 : Logical axis 1 and 2 of the control group (display example: [R1: SLURBT, SLURBT]) 25 : Logical axis of the control group and UNSIGNED CHAR type (display example: [R1: SLURBT, 1]) 27 : Control group and UNSIGNED CHAR type (display example: [R1: 1])
4 to 7	Alarm occurring time (Character strings of 16 letters) Ex.2011/10/10 15:49				
8 to 15	Alarm character strings name (character strings: 32 letters)				It is transmitted in the form of the character strings whose language code was selected by the programming pendant and half- and full-width characters are mixed.
16 to 19	Sub code data additional information character strings (Character strings of 16 letters)				[SV#1] indicates the servo board number 1. [NSU(CPU#1)] indicates that an alarm is found in the function safety unit CPU#1.
20 to 43	Sub code data character strings (Character strings of 96 letters)				
44 to 67	Sub code data character strings reverse display information (Character strings of 96 letters)				Regular characters show [0] and reverse characters show [1]. (display example: [R2S 2])



For the alarm character strings name, it is transmitted in the form of the character strings whose language code was selected by the programming pendant.

Use the same language code as the DX100, or the characters corrupt in case the client side dose not correspond to its language code.

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3.3 Respective Commands for Robot Control

3.3.39 Move instruction command (Type Cartesian coordinates)

Request
Sub header part

<Details>

Command No.	0x8A	
Instance	Specify one out of followings 1:Link absolute position operation 2:Straight absolute position operation 3:Straight increment value operation	Specify the operation number from one to three. 1:Link absolute position operation 2:Straight absolute position operation 3:Straight increment value operation
Attribute	Fixed to "1"	Specify "1".
Service	• Set_Attribute_All: 0x02	Specify the accessing method to the data. 0x02: Write the data to the specified coordinate.

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Specifying control group (Robot)				1 to 8 (Robot No.)
2	Specifying control group (Station)				1 to 24 (Station No.)
3	Specifying the classification in speed				Specify the classification of operations 0: % (Link operation) 1: V (Cartesian operation) 2: VR (Cartesian operation)
4	Specifying a speed				Specify the rate Link operation : 0.01% Cartesian operation V speed : 0.1 mm/s Cartesian operation VR speed : 0.1 degree/s
5	Specifying the operation coordinate				Specify the operation coordinate 16: Base coordinate 17: Robot coordinate 18: User coordinate 19: Tool coordinate
6	X coordinate value (unit: μm)				Refer to following data at the next page for details
7	Y coordinate value (unit: μm)				
8	Z coordinate value (unit: μm)				
9	Tx coordinate value (unit: 0.0001 degree)				
10	Ty coordinate value (unit: 0.0001 degree)				
11	Tz coordinate value (unit: 0.0001 degree)				
12	Reservation				
13	Reservation				
14	Type				
15	Expanded type				
16	Tool No. (0 to 63)				Up to three axes
17	User coordinate No. (1 to 63)				
18	Base 1st axis position (unit: μm)				
19	Base 2nd axis position (unit: μm)				
20	Base 3rd axis position (unit: μm)				
21	Station 1st axis position (pulse value)				
22	Station 2nd axis position (pulse value)				
23	Station 3rd axis position (pulse value)				
24	Station 4th axis position (pulse value)				
25	Station 5th axis position (pulse value)				
26	Station 6th axis position (pulse value)				

Details of data

Please refer to section 3.9.5 “Flip/ No flip” in “DX100 OPERATOR’S MANUAL” prepared for each application.

Form	bit0	0: Front	1: Back	Extended form	bit0	0: $\theta L < 180$,	1: $\theta L \geq 180$
	bit1	0: Upper arm	1: Lower arm		bit1	0: $\theta U < 180$,	1: $\theta U \geq 180$
	bit2	0: Flip	1: No flip		bit2	0: $\theta B < 180$,	1: $\theta B \geq 180$
	bit3	0: $\theta R < 180$,	1: $\theta R \geq 180$		bit3	0: $\theta E < 180$,	1: $\theta E \geq 180$
	bit4	0: $\theta T < 180$,	1: $\theta T \geq 180$		bit4	0: $\theta W < 180$,	1: $\theta W \geq 180$
	bit5	0: $\theta S < 180$,	1: $\theta S \geq 180$		bit5	Reserve	
	bit6	Reserve			bit6	Reserve	
	bit7	Reserve			bit7	Reserve	

To move the base axis, specify the robot No. at the specifying control group, and input the current value to the following coordinate values.



- X coordinate value (unit: μm)
- Y coordinate value (unit: μm)
- Z coordinate value (unit: μm)
- Tx coordinate value (unit: 0.0001 degree)
- Ty coordinate value (unit: 0.0001 degree)
- Tz coordinate value (unit: 0.0001 degree)

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD
Added status	Error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part

3.3.40 Move Instruction Command (Type Pulse)

Request

Sub header part

<Details>

Command No.	0x8B
Instance	Specify one out of followings 1:Link absolute position operation 2:Straight absolute position operation
Attribute	Fixed to “1”
Service	• Set_Attribute_All: 0x02

Specify the operation number from one to three.

1:Link absolute position operation
2:Straight absolute position operation

Specify “1”.

Specify the accessing method to the data.
0x02: Write the data to the specified coordinate.



It is not able to operate the robot and the station at the same time. Setting the both operation at the same time receives the control group setting error (0xB008) from the DX100.

DX100

3 Transmission Procedure

3.3 Respective Commands for Robot Control

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
1	Specifying control group (Robot)				1 to 8 (Robot No.)
2	Specifying control group (Station)				1 to 24 (Station No.)
3	Specifying the classification in speed				Specify the classification of operations 0: % (Link operation) 1: V (Cartesian operation) 2: VR (Cartesian operation)
4	Specifying a speed				Specify the rate Link Operation : 0.01% Cartesian operation V speed : 0.1 mm/s Cartesian operation VR speed : 0.1 degree/s
5	Robot 1st axis pulse value				Up to three axes
6	Robot 2nd axis pulse value				
7	Robot 3rd axis pulse value				
8	Robot 4th axis pulse value				
9	Robot 5thaxis pulse value				
10	Robot 6th axis pulse value				
11	Robot 7th axis pulse value				
12	Robot 8th axis pulse value				
13	Tool No. (0 to 63)				
14	Base 1st axis position (Pulse value)				
15	Base 2nd axis position (Pulse value)				
16	Base 3rdaxis position (Pulse value)				
17	Station 1st axis position (pulse value)				
18	Station 2nd axis position (pulse value)				
19	Station 3rdaxis position (pulse value)				
20	Station 4th axis position (pulse value)				
21	Station 5th axis position (pulse value)				
22	Station 6th axis position (pulse value)				

To move the base axis, specify the robot No. at the specifying control group, and input the each axis value.

- Robot 1st axis pulse value
- Robot 2nd axis pulse value
- Robot 3rd axis pulse value
- Robot 4th axis pulse value
- Robot 5th axis pulse value
- Robot 6th axis pulse value
- Robot 7th axis pulse value
- Robot 8th axis pulse value



Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: not specified • 1: 1 WORD • 2: 2 WORD
Added status	Error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part



It is not able to operate the robot and the station at the same time. Setting the both operation at the same time receives the control group setting error (0xB008) from the DX100.

3.4 File Control Command

Followings are respective commands used in the high-speed Ethernet communication.

Table 3-2: List of File Control Command

No.	Command No.	Instance	Attribute	Service	Command name	Reference
1	0x0	0x0	0x0	0x09	File delete	Refer to <i>section 3.4.1.</i>
2				0x15	File loading command (the PC to the DX100)	Refer to <i>section 3.4.2.</i>
3				0x16	File saving command (the DX100 to the PC)	Refer to <i>section 3.4.3.</i>
4				0x32	File list acquiring command	Refer to <i>section 3.4.4.</i>
5				0x16	File saving command (A batch data backup) (the DX100 to the PC) ¹⁾	Refer to <i>section 3.4.5.</i>

1 This command is available for system software version DS3.00 or higher.

3.4.1 File Deleting Command

Request

Sub header part

<Details>

Command No.	0x0
Instance	0x0
Attribute	0x0
Service	0x09

File deleting process

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
	T	E	S	T
	J	O	B	.
	J	B	I	

<Details>

Specify the job name to be deleted

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part

DX100

3 Transmission Procedure

3.4 File Control Command

3.4.2 File Loading Command

Request

Sub header part

<Details>

Command No.	0x0
Instance	0x0
Attribute	0x0
Service	0x15

File loading process

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
	T	E	S	T
	J	O	B	.
	J	B	I	

<Details>

Specify the job name to be loaded

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part

3.4.3 File Saving Command

Request

Sub header part

<Details>

Command No.	0x0
Instance	0x0
Attribute	0x0
Service	0x16

File saving process

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
	T	E	S	T
	J	O	B	.
	J	B	I	

<Details>

Specify the job names to be saved.

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part

DX100

3 Transmission Procedure

3.4 File Control Command

3.4.4 File List Acquiring Command

Request

Sub header part

<Details>

Command No.	0x0
Instance	0x0
Attribute	0x0
Service	0x32

File list accruing process

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3	<Details>
	*	.	J	B	Refer to "Details of data" for the file type.
	I				

Details of data

No specification	JB I list
.	JB I list
*.JB I	JB I list
*.DAT	DAT file list
*.CND	CND file list
*.PRM	PRM file list
*.SYS	SYS file list
*.LST	LST file list

Answer

Sub header part

<Details>

Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

"1" indicates 1 WORD of added status data, and "2" indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is "1" and that of 2 WORD exists if the code is "2".

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
	1	.	J	B
	I	<CR>	<LF>	2
	2	.	J	B
	I	<CR>	<LF>	3
	3	3	.	J
	B	I	<CR>	<LF>
	T	E	S	T
	0	1	.	J
	B	I	<CR>	<LF>

<Details>
File name + <CR><LF> to input consecutively

3.4.5 File Saving Command (The Batch Data Backup)

Request

Sub header part

<Details>

Command No.	0x0
Instance	0x0
Attribute	0x0
Service	0x16

File saving process

Data part

32bit integer	Byte 0	Byte 1	Byte 2	Byte3
	/	S	P	D
	R	V	/	C
	M	O	S	B
	K	.	B	I
	N			

<Details>

Specify /SPDRV/CMOSBK.BIN

Answer

Sub header part

<Details>

Instance	0x96
Status	Respond by one in the followings • 0x00 : respond normally • Other than 0x00 : respond abnormally
Added status size	• 0: no added status • 1: 1 WORD • 2: 2 WORD
Added status	The error code specified by the added status size

“1” indicates 1 WORD of added status data, and “2” indicates 2 WORD of added status data.

The error code of 1 WORD exists if the added status code is “1” and that of 2 WORD exists if the code is “2”.

Data part

No data part



To set the batch data backup function, set the device as “RAMDISK” as in advance.

It takes about ten minutes to finish backing-up the data by using the batch data backup function.

Refer to *section 2.5 “Setting of a Batch Data Backup Function”* for more detail.

4 Error Code

The following table is the message list of the status.

Added status code	Details
0x08	Requested command is not defined
0x09	The element number of the invalid data is detected
0x1f	An error inherent in vendor occurred (CIP communication protocol corresponds to the vendor specification error) Refer to <i>chapter 5 “Added Status Code”</i> for more details.
0x28	An array number of the requested data does not exist in the specified command.

5 Added Status Code

The following table is the message list of the added status.

Added status code	Details
1010	Command error
1011	Error in number of command operands
1012	Command operand value range over
1013	Command operand length error
1020	Disk full of files
2010	Manipulator operating
2020	Hold by programming pendant
2030	Hold by playback panel
2040	External hold
2050	Command hold
2060	Error/alarm occurring
2070	Servo OFF
2080	Incorrect mode
2090	File accessing by other function
2100	Command remote not set
2110	This data cannot be accessed
2120	This data cannot be loaded
2130	Editing
2150	Running the coordinate conversion function *Refer to the NOTE in the last page of this list.
3010	Turn ON the servo power
3040	Perform home positioning
3050	Confirm positions
3070	Current value not made
3220	Panel lock; mode/cycle prohibit signal is ON
3230	Panel lock; start prohibit signal is ON
3350	User coordinate is not taught
3360	User coordinate is destroyed
3370	Incorrect control group
3380	Incorrect base axis data
3390	Relative job conversion prohibited (at CVTRJ)
3400	Master job call prohibited (parameter)
3410	Master job call prohibited (lamp ON during operation)
3420	Master job call prohibited (teach lock)
3430	Robot calibration data not defined
3450	Servo power cannot be turned ON
3460	Coordinate system cannot be set
4010	Insufficient memory capacity (job registered memory)
4012	Insufficient memory capacity (position data registered memory)
4020	Job editing prohibited
4030	Same job name exists

Added status code	Details
4040	No specified job
4060	Set an execution job
4120	Position data is destroyed
4130	Position data not exist
4140	Incorrect position variable type
4150	END instruction for job which is not master job
4170	Instruction data is destroyed
4190	Invalid character in job name
4200	Invalid character in the label name
4230	Invalid instruction in this system
4420	No step in job to be converted
4430	Already converted
4480	Teach user coordinate
4490	Relative job/ independent control function not permitted
5110	Syntax error (syntax of instruction)
5120	Position data error
5130	No NOP or END
5170	Format error (incorrect format)
5180	Incorrect number of data
5200	Data range over
5310	Syntax error (except instruction)
5340	Error in pseudo instruction specification
5370	Error in condition file data record
5390	Error in JOB data record
5430	System data not same
5480	Incorrect welding function type
6010	The robot/station is under the operation
6020	Not enough memory of the specified device
6030	Cannot be accessed to the specified device
6040	Unexpected auto backup request
6050	CMOS size is over the RAM area
6060	No memory allocation at the power supply on
6070	Accessing error to backup file information
6080	Failed in sorting backup file (Remove)
6090	Failed in sorting backup file (Rename)
6100	Drive name exceeds the specified values
6110	Incorrect device
6120	System error
6130	Auto backup is not available
6140	Cannot be backed up under the auto backup
A000	Undefined command
A001	Instance error
A002	Attribute error
A101	Replying data part size error (hardware limit)
B001	Replying data part size error (software limit)

Added status code	Details
B002	Data use prohibited
B003	Requiring data size error
B004	Outside the data
B005	Data undefined
B006	Specified application unregistered
B007	Specified type unregistered
B008	Control group setting error
B009	Speed setting error
B00A	Operating speed is not setting
B00B	Operation coordinate system setting error
B00C	Type setting error
B00D	Tool No. setting error
B00E	User No. setting error
C001	Address error
C002	System error
C003	System error
C800	System error
CFFF	Other error
D8FA	Transmission exclusive error (BUSY or Semaphore error)
D8F1	Processing the another command (BUSY condition)
E24F	Parameter setting wrong for the system backup
E250	System backup file creating error (confirm if the mode is the remote mode)
E289	System error
E28A	System error
E28B	Disconnect the communication due to receive timeout
E28C	Cannot over write the target file
E29D	System error
E29E	System error
E29F	System error
E2A0	The wrong required pass
E2AA	System error
E2AF	Receive the deletion request of the file that cannot to delete
E2B0	System error
E2B1	The directory cannot to be deleted
E2B2	System error
E2B3	File not found
E2B4	The requested pass is too long
E444	Processing the another command (BUSY condition)
E49D	Format error (data size 0)
E49E	Format error (frame size over)
E49F	Format error (frame size 0)
E4A1	Format error (block number error)
E4A2	Format error (ACK error)

Added status code	Details
E4A3	Format error (processing category error)
E4A4	Format error (access level error)
E4A5	Format error (header size error)
E4A6	Format error (identifier error)
E4A7	Format error (the size of the requested command and received frame are different)
E4A8	System error
E4A9	System error
FFF0	System error
FFF2	System error
FFF3	System error
FFF4	System error
FFF5	System error
FFF6	Too many request and unable to process (BUSY condition)
FFF7	System error
FFF8	System error
FFFE	The remote mode is detected, and disconnect the communication

- Added status code 2150: Running the coordinate conversion function

This error occurs when executes the axis configuration information reading command at the following screen displays.

- Parallel shift job conversion display
- Mirror shift conversion display
- PAM display
- Relative job conversion display
- PMT conversion display
- Position modification display
- Arm bend compensate display
- User coordinate shift display
- Gun teaching position modification display
- 4 point teaching display



It is not only conditions above, but also the error occurs while executing the PMT instruction.

DX100

5 Added Status Code



When the DX100 returning the system error, perform the following procedures.

- 1: Reset the alarm.
- 2: Save the CMOS.BIN, and report the occurrence of the alarm to YASKAWA service representative.

DX100 OPTIONS INSTRUCTIONS

FOR HIGH-SPEED ETHERNET SERVER FUNCTION

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